



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECP2523: DATABASE

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DATABASE DESIGN DESCRIPTIONS: LOGICAL DATABASE DESIGN FOR HASTA VEHICLE RENTAL MANAGEMENT SYSTEM

Group Name: DaTeam

Members:

- 1. MUHAMMAD ADAM ASHRAFF BIN ZAMRI (GROUP LEADER)**
- 2. MIKAEL HAQIMI BIN NAHAR JUNAIDI**
- 3. AHMAD WILDAN BIN MAZANI**
- 4. JOSHUA LING SHANG YANG**

Stakeholder: HASTA TRAVEL & TOURS SDN. BHD

Representative Name	Date Interviewed
1. ALIF FIRDAUS BIN JALALUDIN	4 th JANUARY 2026
2. ALIF FIRDAUS BIN JALALUDIN	8 th JANUARY 2026

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1. Introduction

1.1. Overview about the company

Hasta Travel & Tours Sdn. Bhd. is a licensed service provider for travel and transportation services based in Skudai, Johor. The organization founded by Nabihah Kamarudin alumni of UTM and began as a student movement in August 2018. However, the organization has grown and become a significant contributor to the Universiti Teknologi Malaysia community. Hasta Travel is currently located at the Universiti Teknologi Malaysia Student Mall and provides services in the following areas: car rentals, travel agency services and logistics.

Hasta Travel is designed to accommodate students, lecturers and the general public. They also offer car rental services with their array of vehicles, ranging from economical hatchback cars such as Perodua Axia/Myvi to sedans, SUVs and MPVs.

The mission of Hasta Travel is to facilitate convenient transportation services for UTM students and travelers in Johor by offering cheap and safe transportation. And the vision is to be one of the leading student-centric travel service brands in Malaysia by filling the gap between academic and professional travel services. Their promise is to deliver safety, excellence in vehicle maintenance and support for local university programs.

1.2. The Existing System

Currently, Hasta Travel makes use of both WhatsApp communication and the DriveO platform in managing its bookings and daily activities. The majority of customer interactions, such as booking confirmations, verification of payments and inquiries, are conducted manually by communicating with customers via WhatsApp messages. The staff are required to switch between different platforms in managing their activities.

This method causes various problem such as booking cannot be processed in real time and may increase double booking problems. Next, manual verification may be time-consuming. Lastly, customer information may leak to other resources since it was not saved in a secured database. As the number of customer increasing, the system may overload, crash and be unable to handle it manually.

1.3. The Proposed System

To overcome those problems, we proposed a Vehicle Rental Management System that fully integrates with the real-time booking process. This system will do user registration, identity verification, vehicle availability checking, booking management, payment handling and reporting.

The customers are also able to browse for available vehicles and book the vehicles directly in the system. Employees will be able to access booking information in the staff portal and handle the booking process from start to finish. Employers are able to handle employees information and generate financial reports.

The proposed system will also increase efficiency, minimize errors and enhance security, making it easier for customers and staff to use it. In conclusion, it is believed that the proposed system will allow Hasta Travel to work more professionally.

2. Overview of Project

The HASTA Vehicle Rental Management System project is an initiative to execute the existing manual booking and operations process at HASTA Travel & Tours Sdn. Bhd. The primary goal is to develop a fully integrated digital system that synchronize a real-time operations, reducing human errors such as double bookings, minimizing manual verification time and enhancing customer data security through a secured database.

The system's scope covers end-to-end vehicle rental functions including:

Customer-related processes: User registration, identity verification, browsing available vehicles and direct booking.

Staff management: Handling the booking process from start to complete process via a staff portal.

Management functions: Employers or Data Administrators manage employee information and the generation of essential financial and operational reports.

This document specifically details the foundational database design for the proposed system, outlining the business rules, conceptual and logical Entity-Relationship Diagrams (ERDs), the updated Data Dictionary and the formal Normalization process (up to 3NF). These steps are critical for ensuring the database architecture is organized, efficient and successfully supports the comprehensive functionality of the new system.

Mission and Vision:

- **Mission:** To provide convenient transportation services for UTM students and travelers in Johor by providing cheap and safe transportation services.
- **Vision:** To become one of the leading travel service companies for students in Malaysia by filling the gap between academic and professional travel services.
- **Promise:** The company promises safety, excellence in vehicle maintenance, and support for local university programs.

The Need for a New System (The Existing System's Problems):

- Hasta Travel currently uses WhatsApp communications and the DriveO platform for their bookings. This has resulted in the following problems:
- Real-time bookings cannot be done, which may cause double bookings.
- Verification of customer information also takes time.
- Customer information is also not kept in a secured database, which is a security risk.

The Proposed Solution (HASTA Vehicle Rental Management System):

To solve these problems, a fully integrated Vehicle Rental Management System is recommended. This system is expected to integrate real-time activities, reduce human errors, and strengthen customer information security. The main functions of this system include:

- **Customer Features:** User registration, verification, browsing, and direct booking.
- **Staff Features:** Providing a facility for the staff to view the booking information, thereby enabling them to handle the whole process.
- **Management Features:** Managing employee information, generating important financial reports, etc.

Finally, the recommended system is expected to enhance efficiency, accuracy, and security, enabling Hasta Travel to operate more professionally.

3. Database Conceptual Design

3.1. Updated Business Rule

The following business rules set boundaries on how the Hasta Vehicle Rental Management System can work and how data can be used. These rules are made to limit how data is created, stored and processed in the database.

- **User & Staff Management**
 - A customer must register the account with a unique email address and some personal information(Full Name, Phone Number, IC/Passport) to be able to create the account.
 - A customer must upload personal information, such as a Student Card, Driving License, IC/Passport picture, to change the account status to verified.
 - A staff member is assigned a specific role(Admin/Staff) to determine limitations on the system.
 - A customer can be blacklisted if a staff member notices anomalies from the customer or they have committed violations to prevents from making future bookings.
- **Vehicle Inventory**
 - Each vehicle is identified by a unique license plate number.
 - A vehicle must belong to a specific category(Car or Bike) with a specific rental rate.
 - A vehicle allowed to undergo maintenance handled by staff and all records must be recorded in the vehicle inventory.
- **Booking Operations**
 - A booking must be made by one registered customer only for a specific vehicle.
 - A booking cannot overlap with another booking if overlap staff need to reject the booking and proceed with refunds.
 - Booking can be handled by multiple staff, but needs to change the person in charge of booking management.

- Booking will have inspection records(pre-rental and post-rental) to track vehicle conditions, mileage and fuel levels.
- Payments & Financial
 - Every booking must have individual payment records to track the rental amount and deposit separately.
 - A security deposit will be held and updated to be refunded when the booking is completed or the booking is canceled.
 - A penalty can be issued to the customer if there are any damages or late returns.
 - Penalty records are stored as payment proof once the fine is settled.
- Loyalty & Rewards
 - A customer has a loyalty point tracker to track their current balance points, tier status and total booking counter.
 - Every change of loyalty points to a voucher must be stored in the history table.
 - A voucher can be applied to a booking to provide a discount but one time usability only.
 - Rewards can be categorized(Food or Milestone) and require a specific loyalty tier or booking count to unlock them.

3.2. Conceptual ERD

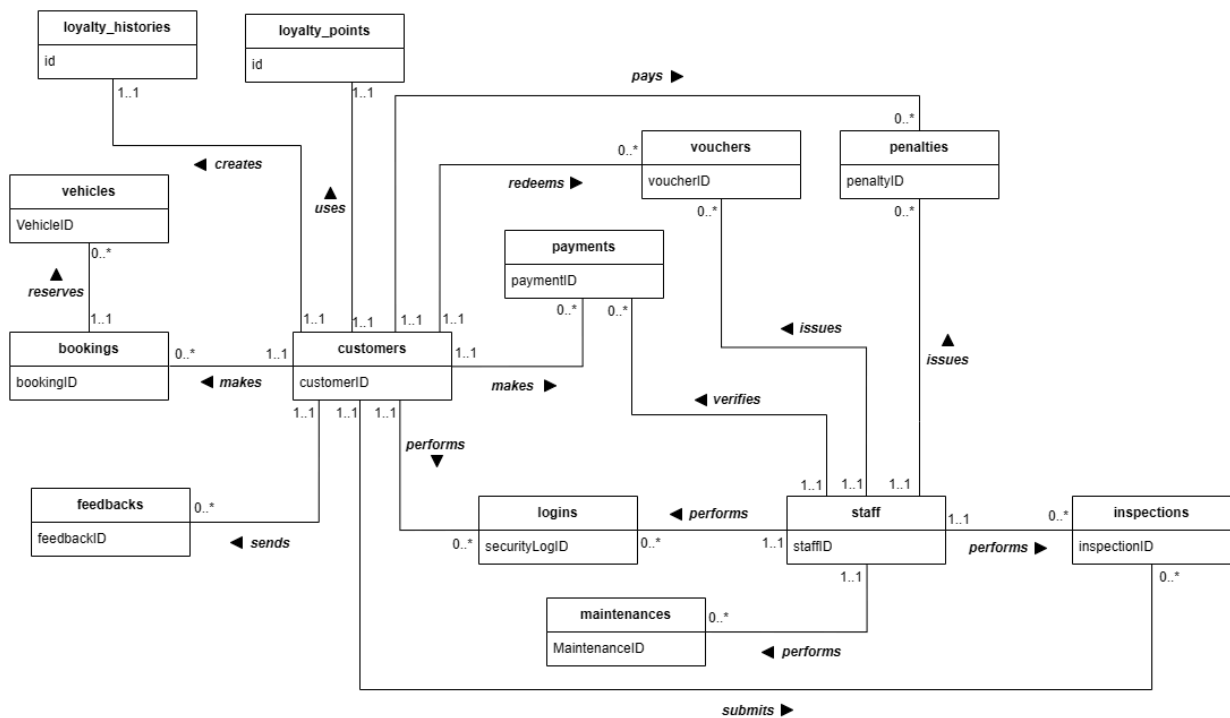


Figure 1: Conceptual ERD of Hasta Vehicle Rental Management System

This Conceptual ERD shows every entity included in our system. And for each entity, there is a unique key, which is the primary key, as shown as in **Figure 3**. There is also the relationship between the entities to show how the entities interact for the development of the entire system.

4. Database Logical Design

4.1. Logical ERD

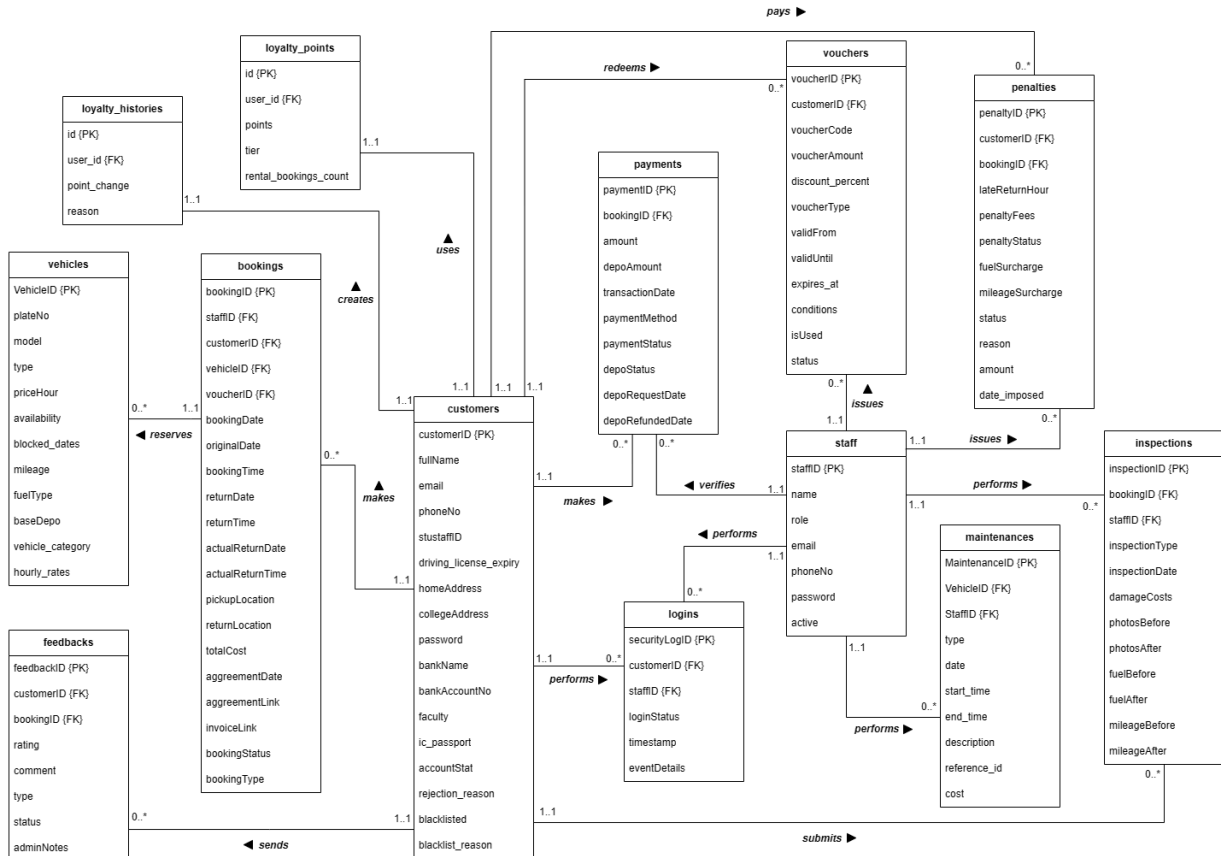


Figure 2: Logical ERD of Hasta Vehicle Rental Management System

The logical ERD is including all the important attributes apart from showing only entities and relationships as the conceptual ERD. **Figure 2** above shows the logical ERD of our system with defining of the primary key and the foreign keys if they exist. For a clearer view, you could click the link below to access this logical ERD:

https://app.diagrams.net/#G10sqDfx9hZOxqegNay3YSM5kcZfN8JE15#%7B%22pageId%22%3A%22PGc3Gn_rrXjI_9cPG3oY%22%7D

4.2. Updated Data Dictionary

The following table, **Table 1** can be used for a comprehensive overview of the system's data architecture with regard to its technical data architecture, which is a translation of the concepts into a database. In this way, it is possible for the system to ensure that all its data, from the identification documents of its customers to the loyalty points accumulated through its automated system, is stored with absolute accuracy through this approach, thus avoiding the risks associated with manual methods of storing data.

Table 1: Updated Data Dictionary based on Logical ERD

Entity Name	Attributes	Data Type & Length	Nulls	Description	Example
staff	staffID (PK)	BIGINT	No	Unique identifier for staff (Primary Key).	1
	name	VARCHAR(100)	No	Full name of the staff member.	Admin User
	role	VARCHAR(50)	No	Access role (e.g., Admin, Staff).	Admin
	email	VARCHAR(100)	No	Unique login email.	admin@dat eam.com
	phoneNo	VARCHAR(20)	Yes	Contact number for operations.	012-34567 89
	password	VARCHAR(255)	No	Hashed security password.	(hashed_str ing)
	active	BOOLEAN	No	Account status flag.	TRUE
customers	customerID (PK)	BIGINT	No	Unique identifier for customer (PK).	1001
	fullName	VARCHAR(100)	No	Full legal name.	Sarah Ali
	email	VARCHAR(100)	No	Unique login email.	sarah@utm .my

Entity Name	Attributes	Data Type & Length	Nulls	Description	Example
	phoneNo	VARCHAR(20)	Yes	Contact number.	019-8765432
	stustaffID	VARCHAR(50)	Yes	Student or Staff Matric ID.	A19EC0001
	driving_licence_expiry	DATE	Yes	Driving license expiration date.	2028-05-20
	ic_passport	VARCHAR(50)	Yes	NRIC or Passport number.	990101-01-5555
	dob	DATE	Yes	Date of Birth.	1999-01-01
	nationality	VARCHAR(50)	Yes	Nationality.	Malaysia
	faculty	VARCHAR(100)	Yes	Faculty or Department name.	Computing
	homeAddress	TEXT	Yes	Permanent home address.	123 Jalan Skudai
	collegeAddress	TEXT	Yes	Residential college address.	Kolej Tun Dr Ismail
	bankName	VARCHAR(100)	Yes	Bank name for refunds.	Maybank
	bankAccount No	VARCHAR(50)	Yes	Bank account number.	164012345678
	emergency_contact_name	VARCHAR(255)	Yes	Name of emergency contact person.	Ali Bin Abu
	emergency_contact_no	VARCHAR(20)	Yes	Phone number of emergency contact.	012-9998888
	accountStat	VARCHAR(50)	No	Account status (Active, Pending).	Active
	previous_account_stat	VARCHAR(255)	Yes	Status held before blacklisting.	Active

Entity Name	Attributes	Data Type & Length	Nulls	Description	Example
	blacklisted	BOOLEAN	No	Flag indicating if user is banned.	FALSE
	blacklist_reason	TEXT	Yes	Reason for blacklisting.	Damaged vehicle
	rejection_reason	TEXT	Yes	Reason for application rejection.	Invalid License
	avatar	VARCHAR(255)	Yes	Path to profile picture file.	profile/1001.jpg
	student_card_image	VARCHAR(255)	Yes	Path to student card image.	docs/card.jpg
	ic_passport_image	VARCHAR(255)	Yes	Path to IC/Passport image.	docs/ic.jpg
	driving_license_image	VARCHAR(255)	Yes	Path to driving license image.	docs/license.jpg
vehicles	VehicleID (PK)	BIGINT	No	Unique identifier for vehicle (PK).	501
	plateNo	VARCHAR(20)	No	Registration plate number (Unique).	JKV 8822
	model	VARCHAR(100)	No	Vehicle model name.	Myvi 1.5 AV
	type	VARCHAR(50)	No	Body type (e.g., Compact, SUV).	Compact
	vehicle_category	VARCHAR(50)	No	Broad category ('car' or 'bike').	car
	brand	VARCHAR(100)	Yes	Manufacturer brand.	Perodua
	year	INTEGER	Yes	Manufacturing year.	2023
	color	VARCHAR(50)	Yes	Vehicle color.	Red

Entity Name	Attributes	Data Type & Length	Nulls	Description	Example
	priceHour	DECIMAL(10,2)	No	Base rental rate per hour.	8.00
	hourly_rates	JSON	Yes	Tiered pricing structure.	{"1": 8, "24": 100}
	availability	BOOLEAN	No	Current availability status.	TRUE
	mileage	INTEGER	No	Current mileage counter.	45000
	owner_name	VARCHAR(100)	Yes	Name of vehicle owner.	Hasta Travel
	owner_phone	VARCHAR(20)	Yes	Contact number of owner.	011-2223333
	owner_nric	VARCHAR(50)	Yes	NRIC/ID of owner.	800101-01-1111
	road_tax_image	VARCHAR(255)	Yes	Path to road tax document.	vehicles/tax.jpg
	grant_image	VARCHAR(255)	Yes	Path to grant document.	vehicles/grant.jpg
	insurance_image	VARCHAR(255)	Yes	Path to insurance document.	vehicles/insurance.jpg
	blocked_dates	JSON	Yes	Dates vehicle is unavailable.	["2025-12-25"]
bookings	bookingID (PK)	BIGINT	No	Unique identifier for booking (PK).	2005
	customerID (FK)	BIGINT	No	Foreign Key to Customer.	1001
	vehicleID (FK)	BIGINT	No	Foreign Key to Vehicle.	501
	staffID (FK)	BIGINT	Yes	Foreign Key to Staff (Approver).	1

Entity Name	Attributes	Data Type & Length	Nulls	Description	Example
	voucherID (FK)	BIGINT	Yes	Foreign Key to Voucher.	801
	bookingDate	DATE	No	Date the booking was created.	2025-12-01
	originalDate	DATE	No	Scheduled start date.	2025-12-05
	bookingTime	TIME	No	Scheduled start time.	10:00:00
	returnDate	DATE	No	Scheduled return date.	2025-12-07
	returnTime	TIME	No	Scheduled return time.	10:00:00
	pickupLocation	VARCHAR(150)	No	Location for vehicle pickup.	KTDI Parking
	returnLocation	VARCHAR(150)	No	Location for vehicle return.	KTDI Parking
	totalCost	DECIMAL(10,2)	No	Total calculated rental cost.	150.00
	bookingStatus	VARCHAR(50)	No	Current status (e.g., Completed).	Completed
	bookingType	VARCHAR(50)	No	Type (Standard, Extended).	Standard
	invoiceLink	VARCHAR(255)	Yes	Path/Link to PDF Invoice.	inv/2005.pdf
	aggreementLink	VARCHAR(255)	Yes	Path/Link to Agreement PDF.	agrmt/2005.pdf
	remarks	TEXT	Yes	Optional customer requests.	Need baby seat
payments	paymentID (PK)	BIGINT	No	Unique identifier for payment (PK).	3001

Entity Name	Attributes	Data Type & Length	Nulls	Description	Example
	bookingID (FK)	BIGINT	Yes	FK to Booking (Nullable).	2005
	amount	DECIMAL(10,2)	No	Amount paid in transaction.	150.00
	depoAmount	DECIMAL(10,2)	No	Security deposit amount held.	50.00
	transactionDate	DATETIME	No	Date and time of payment.	2025-12-01 14:30
	paymentMethod	VARCHAR(50)	No	Method (e.g., QR, Cash).	DuitNow
	paymentStatus	VARCHAR(50)	No	Status (e.g., Paid, Pending).	Paid
	depoStatus	VARCHAR(50)	No	Deposit status (Held/Refunded).	Refunded
penalties	penaltyID (PK)	BIGINT	No	Unique identifier for penalty (PK).	4001
	bookingID (FK)	BIGINT	Yes	FK to Booking (Nullable).	2005
	customerID (FK)	BIGINT	Yes	FK to Customer.	1001
	amount	DECIMAL(10,2)	No	Total penalty amount.	20.00
	reason	TEXT	Yes	Reason for the penalty.	Late Return
	date_imposed	DATETIME	Yes	Date/Time penalty was issued.	2025-12-07 12:00
	lateReturnHour	INTEGER	No	Hours returned late (if any).	2
	penaltyFees	DECIMAL(10,2)	No	Specific fee for lateness.	20.00

Entity Name	Attributes	Data Type & Length	Nulls	Description	Example
	penaltyStatus	VARCHAR(50)	No	Payment status (Unpaid/Paid).	Unpaid
	payment_proof	VARCHAR(255)	Yes	Path to receipt image.	penalties/receipt.jpg
	paid_at	TIMESTAMP	Yes	Timestamp when paid.	2026-01-08 10:00
maintenance	MaintenanceID (PK)	BIGINT	No	Unique identifier (PK).	901
	VehicleID (FK)	BIGINT	No	FK to Vehicle being serviced.	501
	StaffID (FK)	BIGINT	Yes	FK to Staff overseeing work.	1
	description	VARCHAR(255)	Yes	Details of maintenance work.	Oil Change
	cost	DECIMAL(10,2)	No	Cost incurred.	120.00
	date	DATE	Yes	General date of service.	2025-12-20
	start_time	DATETIME	Yes	Precise start time.	2025-12-20 09:00
	end_time	DATETIME	Yes	Precise completion time.	2025-12-20 11:00
	type	VARCHAR(255)	No	Type (maintenance, holiday, etc).	maintenance
	reference_id	VARCHAR(255)	Yes	External reference ID.	REF-12345
vouchers	voucherID (PK)	BIGINT	No	Unique identifier (PK).	801
	customerID (FK)	BIGINT	Yes	Foreign Key to Customer.	1001

Entity Name	Attributes	Data Type & Length	Nulls	Description	Example
	user_id (FK)	BIGINT	Yes	Alias for customerID (FK).	1001
	voucherCode	VARCHAR(50)	No	Unique code for redemption.	SDKTN-8821
	code	VARCHAR(255)	Yes	Alias for voucherCode.	SDKTN-8821
	voucherAmount	DECIMAL(10,2)	No	Fixed discount amount.	5.00
	discount_percent	INTEGER	Yes	Percentage discount value.	20
	voucherType	VARCHAR(50)	No	Type (Rental Discount, Merchant).	Merchant Reward
	validFrom	DATE	No	Start of validity period.	2026-01-01
	validUntil	DATE	No	End of validity period.	2026-03-01
	expires_at	DATE	Yes	Alias for validUntil.	2026-03-01
	conditions	TEXT	Yes	Terms of use.	Min spend RM50
	terms_conditions	TEXT	Yes	Alias for conditions.	Min spend RM50
	isUsed	BOOLEAN	No	Usage flag.	FALSE
	status	VARCHAR(255)	No	Status string (e.g. 'unused').	unused
	redeem_place	VARCHAR(255)	Yes	Location to redeem.	Sedap Kitchen
loyalty_points	id (PK)	BIGINT	No	Unique identifier (PK).	601
	user_id (FK)	BIGINT	No	Foreign Key to Customer.	1001

Entity Name	Attributes	Data Type & Length	Nulls	Description	Example
	points	INTEGER	No	Current points balance.	500
	tier	VARCHAR(50)	No	Tier (Bronze, Silver, Gold).	Gold
	rental_bookings_count	INTEGER	No	Qualified bookings count.	5
loyalty_histories	id (PK)	BIGINT	No	Unique identifier (PK).	950
	user_id (FK)	BIGINT	No	Foreign Key to Customer.	1001
	points_change	INTEGER	No	Points added (+) or deducted (-).	+100
	reason	VARCHAR(255)	No	Transaction description.	Rental Reward
rewards	id (PK)	BIGINT	No	Unique identifier (PK).	701
	name	VARCHAR(255)	No	Name of the reward.	SEDAP KITCHEN
	offer_description	VARCHAR(255)	No	Details of the offer.	RM 5 OFF
	points_required	INTEGER	No	Points cost to redeem.	200
	category	VARCHAR(50)	No	Category (Food, Milestone).	Food
	code_prefix	VARCHAR(255)	No	Prefix for voucher codes.	SDKTN
	milestone_step	INTEGER	Yes	Booking count trigger step.	12
	discount_percent	INTEGER	No	Discount percentage value.	50

Entity Name	Attributes	Data Type & Length	Nulls	Description	Example
inspections	inspectionID (PK)	BIGINT	No	Unique identifier (PK).	1101
	bookingID (FK)	BIGINT	No	FK to Booking.	2005
	staffID (FK)	BIGINT	Yes	FK to Staff.	1
	inspectionType	VARCHAR(50)	No	Pre-rental or Post-rental.	Post-rental
	damageCosts	DECIMAL(10,2)	No	Estimated cost of damages.	0.00
	photosAfter	TEXT	Yes	Photos after rental (JSON/Path).	["img1.jpg"]
	photosBefore	TEXT	Yes	Photos before rental (JSON/Path).	["img0.jpg"]
feedback	feedbackID (PK)	BIGINT	No	Unique identifier (PK).	1201
	customerID (FK)	BIGINT	No	Foreign Key to Customer.	1001
	bookingID (FK)	BIGINT	Yes	Foreign Key to Booking.	2005
	rating	INTEGER	No	Rating score (e.g., 1-5).	5
	comment	TEXT	Yes	Textual feedback.	Great service!
	type	VARCHAR(50)	No	Feedback type (Complaint/Review).	Review
	status	VARCHAR(50)	No	Status (Pending/Reviewed).	Reviewed

4.3. Normalization

4.3.1 First Normal Form (1NF)

1NF ensures that all attributes are atomic and there are no repeating groups. The document defines the initial tables, implicitly satisfying 1NF by valued attributes in **VEHICLES** through new tables which are **VEHICLE_BLOCKED_DATES** and **VEHICLE_RATES**. **The Problem:** The **VEHICLES** table originally stored **blocked_dates** and **hourly_rates** as **JSON objects**, which violates the 1NF rule that every column must contain an indivisible (atomic) value.

VEHICLES (vehicleID PK, plateNo, model, type, priceHour, availability, blocked_dates, mileage, fuelType, baseDepo, vehicle_category, brand, year, color, hourly_rates, owner_name, owner_phone, owner_nric, image, road_tax_image, grant_image, insurance_image, created_at, updated_at)

The 1NF Solution: These were decomposed into the following atomic schemas:

VEHICLE_BLOCKED_DATES (id PK, vehicleID FK, blocked_date)

VEHICLE_RATES (id PK, vehicleID FK, rate_type, price)

*Note that with this justification, **other tables are already in 1NF**.

4.3.2 Second Normal Form (2NF)

While **all tables** in the Hasta Vehicle Rental system **satisfy 2NF** due to the use of **single-column primary keys**, the **2NF stage specifically focuses on ensuring that multi-entity relationships such as those found in the Loyalty, Vouchers, Penalties and Rewards modules are properly established through Foreign Key linkages**. This ensures that every non-key attribute is fully functionally dependent on its respective primary identifier while maintaining referential integrity across the database.

- **LOYALTY_POINTS** (id PK, **user_id** FK, points, tier, rental_bookings_count, created_at, updated_at)
- **LOYALTY_HISTORIES** (id PK, **user_id** FK, points_change, reason, created_at, updated_at)
- **VOUCHERS** (voucherID PK, **customerID** FK, **user_id** FK, voucherCode, code, voucherAmount, discount_percent, status, isUsed, created_at, updated_at)
- **PENALTIES** (penaltyID PK, **customerID** FK, **bookingID** FK, amount, reason, penaltyStatus, paid_at, created_at)
- **REWARD_CATEGORIES** (category PK, icon_class, color_class)
- **REWARDS** (**id** PK, name, offer_description, points_required, code_prefix, validity_months, category FK, is_active, milestone_step, discount_percent, created_at, updated_at)

*Note that with this justification, **other tables are already in 2NF**

4.3.3 Third Normal Form (3NF)

3NF requires the table to be in 2NF and eliminates transitive dependencies, meaning no non-key attribute is dependent on another non-key attribute. This is achieved by splitting the large **CUSTOMERS** and **VEHICLES** tables into smaller and related entities.

Decomposition for 3NF:

From **CUSTOMERS** (to remove transitive dependencies):

The original **CUSTOMERS** table contained personal information, addresses, bank details, emergency contacts and document links. These attributes are only transitively dependent on **customerID** via the attributes themselves (e.g., **bankAccountNo** depends on the bank details, which in turn depends on the **customerID**).

The table is decomposed into:

- **CUSTOMER_ADDRESSES** (removes **homeAddress**, **collegeAddress**)
- **CUSTOMER_DOCUMENTS** (removes **driving_license_expiry**, **student_card_image**, **ic_passport_image**, **driving_license_image**)
- **BANK_DETAILS** (removes **bankName**, **bankAccountNo**)
- **EMERGENCY_CONTACTS** (removes **emergency_contact_name**, **emergency_contact_no**)
- **REJECTION_LOGS** (removes **rejection_reason**, **blacklist_reason**)

From **VEHICLES** (to remove transitive dependencies):

The original **VEHICLES** table included details about the vehicle owner and registration documents. These attributes (owner details, document paths) describe the vehicle owner/documents not the vehicle itself and are transitively dependent on **vehicleID**.

The table is decomposed into:

- **VEHICLE_OWNERS** (removes **owner_name**, **owner_phone**, **owner_nric**)
- **VEHICLE_DOCS** (removes **road_tax_image**, **grant_image**, **insurance_image**)

*Note that with this justification, **other tables** are **already in 3NF**

5. Relational Database Schemas (after normalization)

The final schemas after normalization:

5.1 User & Staff Module

STAFF (staffID PK, name, role, email, phoneNo, password, active, created_at, updated_at)

CUSTOMERS (customerID PK, fullName, email, phoneNo, stustaffID, password, faculty, ic_passport, nationality, dob, avatar, accountStat, previous_account_stat, blacklisted, created_at, updated_at)

CUSTOMER_ADDRESSES (addressID PK, customerID FK, type, address_detail)

CUSTOMER_DOCUMENTS (docID PK, customerID FK, driving_license_expiry, student_card_image, ic_passport_image, driving_license_image)

BANK_DETAILS (bankID PK, customerID FK, bankName, bankAccountNo)

EMERGENCY_CONTACTS (contactID PK, customerID FK, contact_name, contact_no)

REJECTION_LOGS (rejectionID PK, customerID FK, rejection_reason, blacklist_reason)

5.2 Vehicle & Maintenance Module

VEHICLES (vehicleID PK, plateNo, model, type, priceHour, availability, mileage, fuelType, baseDepo, vehicle_category, brand, year, color, image, created_at, updated_at)

VEHICLE_OWNERS (ownerID PK, vehicleID FK, owner_name, owner_phone, owner_nric)

VEHICLE_DOCS (vdocID PK, vehicleID FK, road_tax_image, grant_image, insurance_image)

MAINTENANCE (maintenanceID PK, vehicleID FK, staffID FK, type, date, start_time, end_time, description, reference_id, cost, created_at, updated_at)

5.3 Booking & Operations Module

BOOKINGS (bookingID PK, staffID FK, customerID FK, vehicleID FK, voucherID FK, bookingDate, originalDate, bookingTime, returnDate, returnTime, actualReturnDate, actualReturnTime, pickupLocation, returnLocation, totalCost, agreementDate, agreementLink, invoiceLink, remarks, bookingStatus, bookingType, created_at, updated_at)

INSPECTIONS (inspectionID PK, bookingID FK, staffID FK, inspectionType, inspectionDate, damageCosts, photosBefore, photosAfter, fuelBefore, fuelAfter, mileageBefore, mileageAfter, created_at, updated_at)

5.4 Finance & Loyalty Module

PAYMENTS (paymentID PK, bookingID FK, amount, depoAmount, transactionDate, paymentMethod, paymentStatus, depoStatus, depoRequestDate, depoRefundedDate, isInstallment, installmentDetails, created_at, updated_at)

PENALTIES (penaltyID PK, customerID FK, bookingID FK, lateReturnHour, penaltyFees, penaltyStatus, fuelSurcharge, mileageSurcharge, status, reason, amount, payment_proof, paid_at, date_imposed, created_at, updated_at)

VOUCHERS (voucherID PK, customerID FK, user_id FK, voucherCode, code, voucherAmount, discount_percent, voucherType, redeem_place, validFrom, validUntil, expires_at, conditions, terms_conditions, isUsed, status, created_at, updated_at)

LOYALTY_POINTS (id PK, user_id FK, points, tier, rental_bookings_count, created_at, updated_at)

LOYALTY_HISTORIES (id PK, user_id FK, points_change, reason, created_at, updated_at)

REWARDS (id PK, name, offer_description, points_required, code_prefix, validity_months, category FK, is_active, milestone_step, discount_percent, created_at, updated_at)

REWARD_CATEGORIES (category PK, icon_class, color_class)

6. SQL Statements (DDL & DML)

6.1 SQL Statements (DDL)

6.1.1 System & Utility Tables

```
CREATE TABLE `sessions` (  
  `id` VARCHAR(100) PRIMARY KEY,  
  `user_id` BIGINT UNSIGNED NULL,  
  `ip_address` VARCHAR(45) NULL,  
  `user_agent` TEXT NULL,  
  `payload` LONGTEXT NOT NULL,  
  `last_activity` INT NOT NULL,  
  INDEX (`user_id`),  
  INDEX (`last_activity`)  
);
```

```
CREATE TABLE `cache` (  
  `key` VARCHAR(100) PRIMARY KEY,  
  `value` MEDIUMTEXT NOT NULL,  
  `expiration` INT NOT NULL  
);
```

```
CREATE TABLE `cache_locks` (  
  `key` VARCHAR(100) PRIMARY KEY,  
  `owner` VARCHAR(100) NOT NULL,  
  `expiration` INT NOT NULL  
);
```

```
CREATE TABLE `jobs` (  
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
  `queue` VARCHAR(100) NOT NULL,  
  `payload` LONGTEXT NOT NULL,  
  `attempts` TINYINT UNSIGNED NOT NULL,  
  `reserved_at` INT UNSIGNED NULL,  
  `available_at` INT UNSIGNED NOT NULL,  
  `created_at` INT UNSIGNED NOT NULL,  
  INDEX (`queue`)  
);
```

```
CREATE TABLE `job_batches` (  
  `id` VARCHAR(100) PRIMARY KEY,  
  `name` VARCHAR(100) NOT NULL,  
  `total_jobs` INT NOT NULL,
```



```

`pending_jobs` INT NOT NULL,
`failed_jobs` INT NOT NULL,
`failed_job_ids` LONGTEXT NOT NULL,
`options` MEDIUMTEXT NULL,
`cancelled_at` INT NULL,

`created_at` INT NOT NULL,
`finished_at` INT NULL
);

```

```

CREATE TABLE `failed_jobs` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `uuid` VARCHAR(100) UNIQUE NOT NULL,
  `connection` TEXT NOT NULL,
  `queue` TEXT NOT NULL,
  `payload` LONGTEXT NOT NULL,
  `exception` LONGTEXT NOT NULL,
  `failed_at` TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

```

```

CREATE TABLE `notifications` (
  `id` CHAR(36) PRIMARY KEY,
  `type` VARCHAR(255) NOT NULL,
  `notifiable_type` VARCHAR(255) NOT NULL,
  `notifiable_id` BIGINT UNSIGNED NOT NULL,
  `data` TEXT NOT NULL,
  `read_at` TIMESTAMP NULL,
  `created_at` TIMESTAMP NULL,
  `updated_at` TIMESTAMP NULL,
  INDEX (`notifiable_type`, `notifiable_id`)
);

```

6.1.2 Logins Table

```

CREATE TABLE `logins` (
  `securityLogID` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `customerID` BIGINT UNSIGNED NULL,
  `staffID` BIGINT UNSIGNED NULL,
  `loginStatus` VARCHAR(50) NOT NULL,
  `timestamp` DATETIME NOT NULL,
  `eventDetails` TEXT NULL,

  `created_at` TIMESTAMP NULL,
  `updated_at` TIMESTAMP NULL,

```

```

FOREIGN KEY (`customerID`) REFERENCES `customers`(`customerID`) ON
DELETE CASCADE,
FOREIGN KEY (`staffID`) REFERENCES `staff`(`staffID`) ON DELETE
CASCADE
);

```

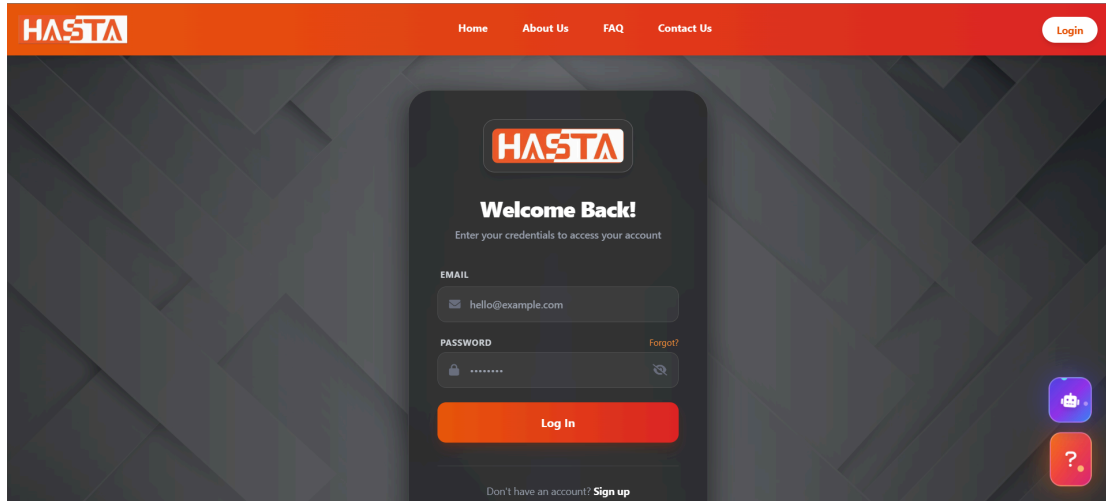


Figure 3: Login interface (Customer)

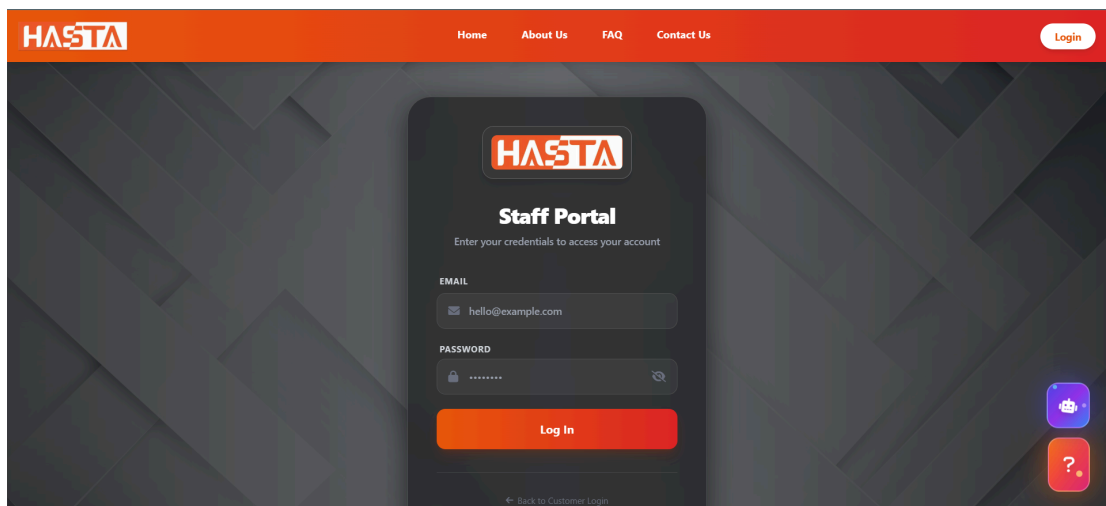


Figure 4: Login interface (Staff)

We are using two different interfaces for both different users, where customers will login through interface in **Figure 3** to access vehicle rental system while staff will login through interface in **Figure 4** to access system management page. They are required to fill in their email and password, those same as the sign up phase.

6.1.3 Customer Table

```

CREATE TABLE `customers` (
  (Profile Details)
  `customerID` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `fullName` VARCHAR(100) NOT NULL,
  `email` VARCHAR(100) UNIQUE NOT NULL,

```

`phoneNo` VARCHAR(20) NULL,
`stustaffID` VARCHAR(50) NULL,
`driving\_license\_expiry` DATE NULL,
`homeAddress` TEXT NULL,
`collegeAddress` TEXT NULL,
`password` VARCHAR(255) NOT NULL,

(Banking)

`bankName` VARCHAR(100) NULL,
`bankAccountNo` VARCHAR(50) NULL,

(Enhanced Profile)

`faculty` VARCHAR(100) NULL,
`ic\_passport` VARCHAR(50) NULL,
`nationality` VARCHAR(50) NULL,
`dob` DATE NULL,
`emergency\_contact\_no` VARCHAR(20) NULL,
`emergency\_contact\_name` VARCHAR(255) NULL,

(Images)

`avatar` VARCHAR(255) NULL,
`student\_card\_image` VARCHAR(255) NULL,
`ic\_passport\_image` VARCHAR(255) NULL,
`driving\_license\_image` VARCHAR(255) NULL,

(Status & Security)

`accountStat` VARCHAR(50) DEFAULT 'active',
`previous\_account\_stat` VARCHAR(255) NULL,
`rejection\_reason` TEXT NULL,
`blacklisted` BOOLEAN DEFAULT FALSE,
`blacklist\_reason` TEXT NULL,

`created\_at` TIMESTAMP NULL,
`updated\_at` TIMESTAMP NULL

);

Figure 5: Customer Profile interface

Figure 5 shows the interface of the customer profile, where it will show all the personal information, documentations and address and bank details that could be either filled in or updated by the customers, then will wait for verification from staff after saving the latest information.

6.1.4 Staff Table

```
CREATE TABLE `staff` (
  `staffid` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `name` VARCHAR(100) NOT NULL,
  `role` VARCHAR(50) DEFAULT 'staff',
  `email` VARCHAR(100) UNIQUE NOT NULL,
  `phoneNo` VARCHAR(20) NULL,
  `password` VARCHAR(255) NOT NULL,
  `active` BOOLEAN DEFAULT TRUE,

  `created_at` TIMESTAMP NULL,
  `updated_at` TIMESTAMP NULL
);
```

Figure 6: Staff Profile interface

Figure 6 shows the interface of the staff profile where it will display the basic information for example full name, email address and phone number. They could also change a new password if they want at the section below the personal information.

6.1.5 Vehicle Table

```
CREATE TABLE `vehicles` (  
  (Vehicle Details)  
  `VehicleID` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
  `plateNo` VARCHAR(20) UNIQUE NOT NULL,  
  `model` VARCHAR(100) NOT NULL,  
  `type` VARCHAR(50) NOT NULL,  
  `priceHour` DECIMAL(10, 2) NOT NULL,  
  `availability` BOOLEAN DEFAULT TRUE,  
  `mileage` INT NOT NULL,  
  `fuelType` VARCHAR(50) NOT NULL,  
  `baseDepo` DECIMAL(10, 2) NOT NULL,  
  
  (Enhanced Vehicle Details)  
  `vehicle_category` VARCHAR(50) DEFAULT 'car',  
  `brand` VARCHAR(100) NULL,  
  `year` INT NULL,  
  `color` VARCHAR(50) NULL,  
  `hourly_rates` JSON NULL,  
  `blocked_dates` JSON NULL,  
  
  (Owner Info)  
  `owner_name` VARCHAR(100) NULL,  
  `owner_phone` VARCHAR(20) NULL,  
  `owner_nric` VARCHAR(50) NULL,  
  
  (Documentation Images)  
  `image` VARCHAR(255) NULL,  
  `road_tax_image` VARCHAR(255) NULL,  
  `grant_image` VARCHAR(255) NULL,  
  `insurance_image` VARCHAR(255) NULL,  
  
  `created_at` TIMESTAMP NULL,  
  `updated_at` TIMESTAMP NULL  
);
```

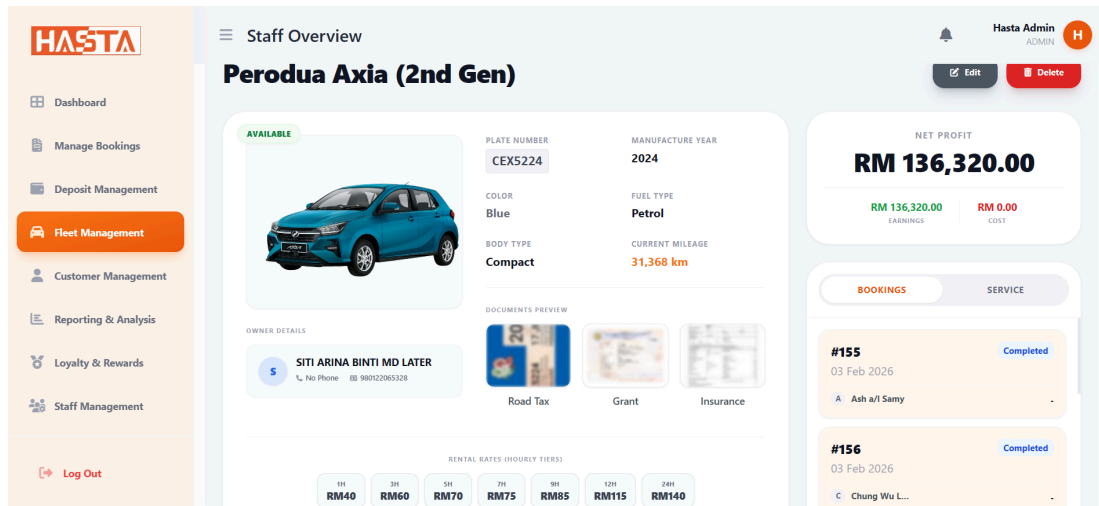


Figure 7: Vehicle Information interface

In **Figure 7**, it shows the details about the vehicle of the selected vehicle, staff could update the information at any time. Staff could also upload the vehicle documents like road tax, grant and insurance so they could easily preview those documents through this interface.

6.1.6 Vehicle Maintenance Table

```
CREATE TABLE `maintenances` (
  `MaintenanceID` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `VehicleID` BIGINT UNSIGNED NOT NULL,
  `StaffID` BIGINT UNSIGNED NULL,
  `date` DATE NULL,
  `start_time` DATETIME NULL,
  `end_time` DATETIME NULL,
  `type` VARCHAR(255) DEFAULT 'maintenance', -- maintenance, holiday, delivery
  `description` VARCHAR(255) NULL,
  `reference_id` VARCHAR(255) NULL,
  `cost` DECIMAL(10, 2) DEFAULT 0.00,
  `created_at` TIMESTAMP NULL,
  `updated_at` TIMESTAMP NULL,

  FOREIGN KEY (`VehicleID`) REFERENCES `vehicles` (`VehicleID`) ON DELETE CASCADE,
  FOREIGN KEY (`StaffID`) REFERENCES `staff` (`staffID`) ON DELETE SET NULL
);
```

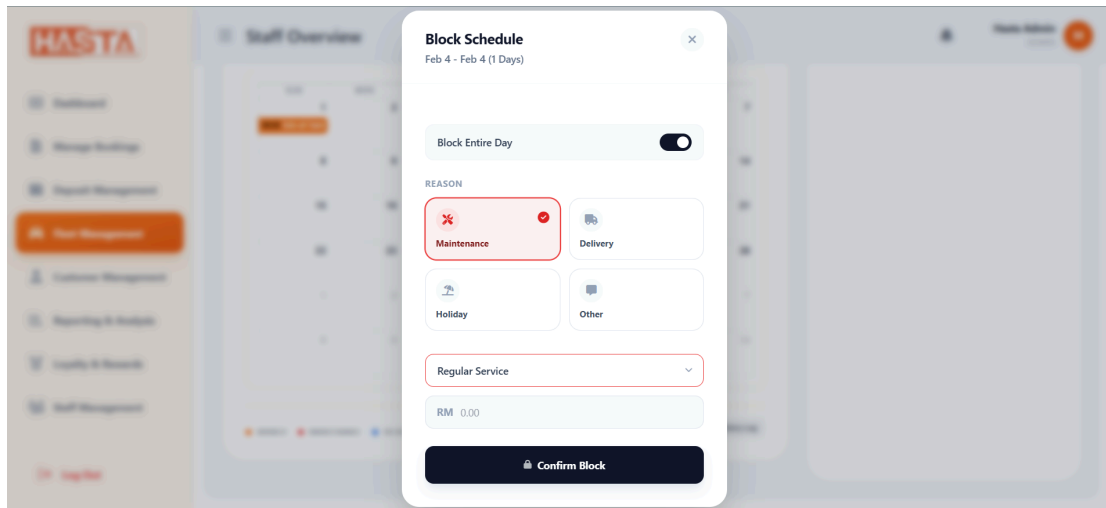


Figure 8: Add Maintenance to Entire Vehicle interface

In this interface as shown in **Figure 8**, staff could add a block schedule for the entire vehicle for either maintenance, delivery, holiday or others. At the maintenance facility, staff can choose the type of maintenance like regular service, tire change, battery replacement, major repair and inspection with their amount of cost.

6.1.7 Bookings Table

```
CREATE TABLE `bookings` (
  `bookingID` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `staffID` BIGINT UNSIGNED NULL,
  `customerID` BIGINT UNSIGNED NOT NULL,
  `vehicleID` BIGINT UNSIGNED NOT NULL,
  `voucherID` BIGINT UNSIGNED NULL,

  `bookingDate` DATE NOT NULL,
  `originalDate` DATE NOT NULL,
  `bookingTime` TIME NOT NULL,

  `returnDate` DATE NOT NULL,
  `returnTime` TIME NOT NULL,
  `actualReturnDate` DATE NULL,
  `actualReturnTime` TIME NULL,

  `pickupLocation` VARCHAR(150) NOT NULL,
  `returnLocation` VARCHAR(150) NOT NULL,
  `totalCost` DECIMAL(10, 2) NOT NULL,

  `aggreementDate` DATE NULL,
  `aggreementLink` VARCHAR(255) NULL,
  `invoiceLink` VARCHAR(255) NULL, --
  `remarks` TEXT NULL, --
```

```

`bookingStatus` VARCHAR(50) DEFAULT 'Pending',
`bookingType` VARCHAR(50) DEFAULT 'Standard',

`created_at` TIMESTAMP NULL,
`updated_at` TIMESTAMP NULL,

FOREIGN KEY (`staffID`) REFERENCES `staff` (`staffID`),
FOREIGN KEY (`customerID`) REFERENCES `customers` (`customerID`),
FOREIGN KEY (`vehicleID`) REFERENCES `vehicles` (`VehicleID`),
FOREIGN KEY (`voucherID`) REFERENCES `vouchers` (`voucherID`)
);

```

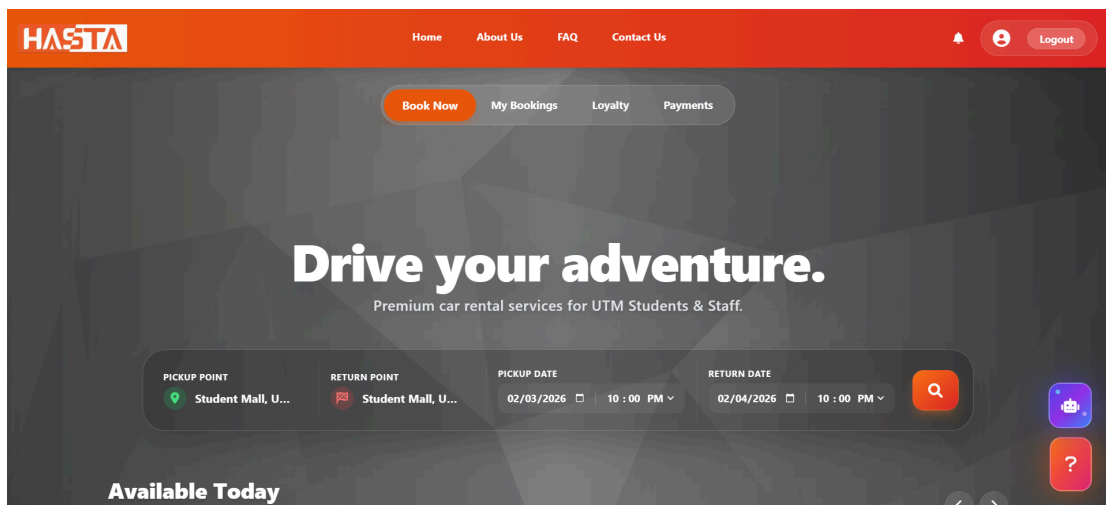


Figure 9: Book Now interface

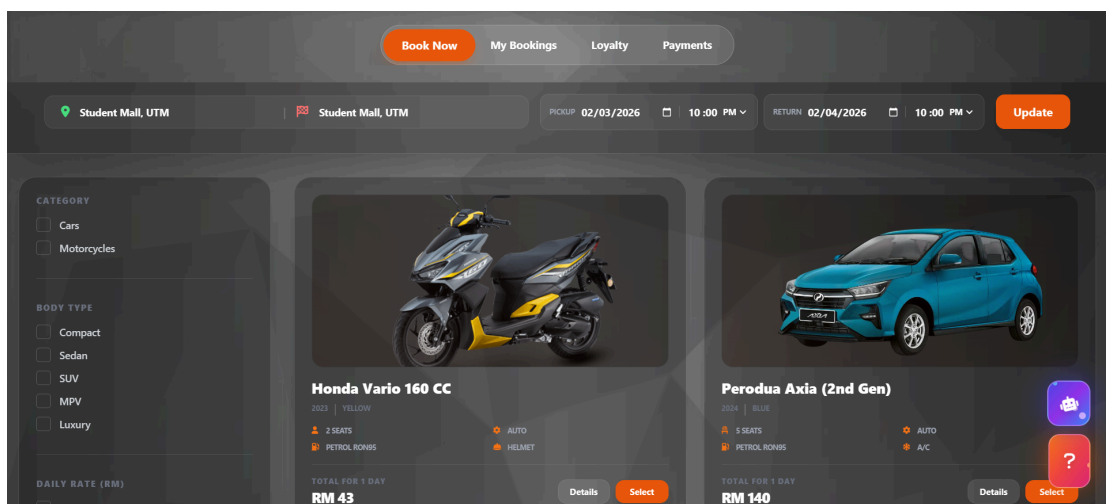


Figure 10: Vehicle Selection interface

If a customer wants to search for a vehicle they prefer, they can choose the pickup and return locations, date and time with their preference as shown in **Figure 9**. After clicking search, the customer will be directed to the vehicle selection stage as shown in **Figure 10**. The customer could select the car they prefer.

6.1.8 Payments Table

```
CREATE TABLE `payments` (  
  `paymentID` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
  `bookingID` BIGINT UNSIGNED NULL,  
  
  `amount` DECIMAL(10, 2) NOT NULL,  
  `depoAmount` DECIMAL(10, 2) NOT NULL,  
  `transactionDate` DATETIME NOT NULL,  
  `paymentMethod` VARCHAR(50) NOT NULL,  
  `paymentStatus` VARCHAR(50) NOT NULL,  
  
  `depoStatus` VARCHAR(50) DEFAULT 'Pending',  
  `depoRequestDate` DATE NULL,  
  `depoRefundedDate` DATE NULL,  
  
  `isInstallment` BOOLEAN DEFAULT FALSE,  
  `installmentDetails` TEXT NULL,  
  
  `created_at` TIMESTAMP NULL,  
  `updated_at` TIMESTAMP NULL,  
  
  FOREIGN KEY (`bookingID`) REFERENCES `bookings`(`bookingID`) ON  
  DELETE CASCADE  
);
```

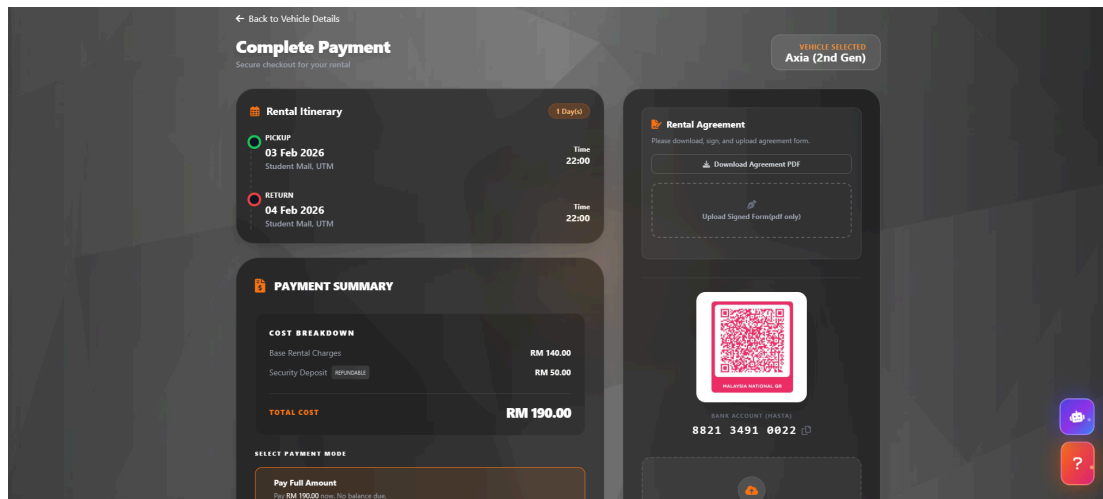


Figure 11: Payment interface

Figure 11 shows the payment interface after selection of the vehicle. Customers will need to sign for the agreement, make payment via QR code and upload the receipt for the references.

6.1.9 Penalty Table

```
CREATE TABLE `penalties` (  

```

```

`penaltyID` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
`customerID` BIGINT UNSIGNED NULL,
`bookingID` BIGINT UNSIGNED NULL,

`reason` TEXT NULL,
`amount` DECIMAL(10, 2) DEFAULT 0.00,
`date_imposed` DATETIME NULL, -- Converted to DATETIME

`payment_proof` VARCHAR(255) NULL,
`paid_at` TIMESTAMP NULL,

`lateReturnHour` INT DEFAULT 0,
`penaltyFees` DECIMAL(10, 2) DEFAULT 0.00,
`penaltyStatus` VARCHAR(50) DEFAULT 'Unpaid',
`fuelSurcharge` DECIMAL(10, 2) DEFAULT 0.00,
`mileageSurcharge` DECIMAL(10, 2) DEFAULT 0.00,
`status` VARCHAR(50) DEFAULT 'Pending',

`created_at` TIMESTAMP NULL,
`updated_at` TIMESTAMP NULL,

FOREIGN KEY (`bookingID`) REFERENCES `bookings` (`bookingID`) ON
DELETE CASCADE,
FOREIGN KEY (`customerID`) REFERENCES `customers` (`customerID`) ON
DELETE CASCADE
);

```

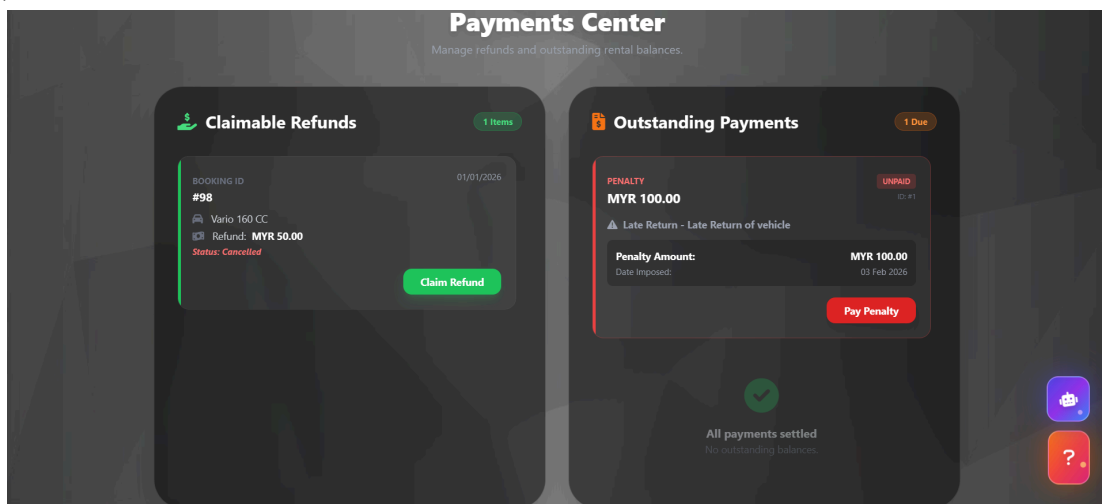


Figure 12: Payment Center interface

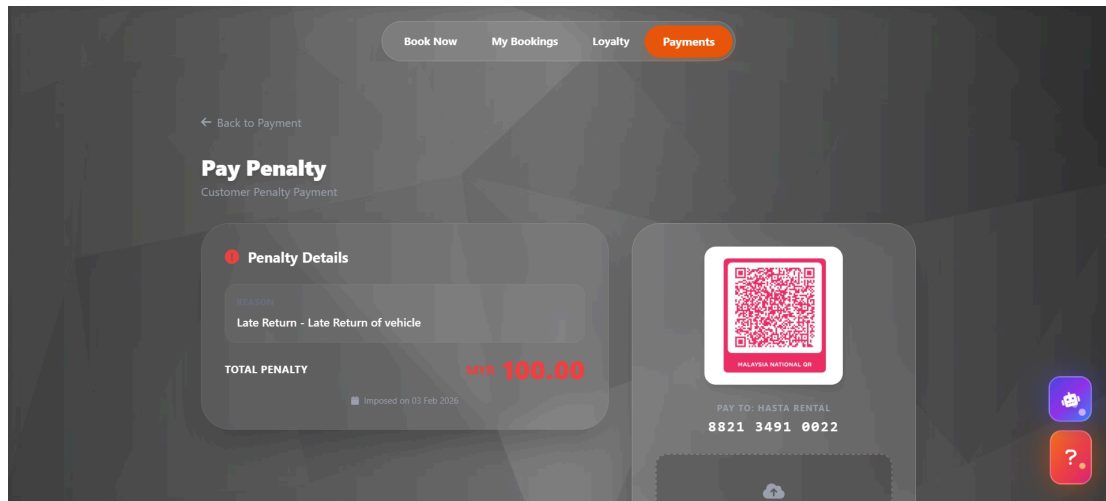


Figure 13: Penalty interface

If a customer has any penalty, it will be shown in the interface as shown in **Figure 12**. Then customers will be required to finish the penalty first before making further bookings. The customer could pay for the penalty in the interface as shown in **Figure 13**.

6.1.10 Voucher Table

```
CREATE TABLE `vouchers` (
  `voucherID` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `customerID` BIGINT UNSIGNED NULL,
  `user_id` BIGINT UNSIGNED NULL,
  `voucherCode` VARCHAR(50) UNIQUE NOT NULL,
  `code` VARCHAR(255) NULL,
  `voucherAmount` DECIMAL(10, 2) NOT NULL,
  `discount_percent` INT NULL,
  `voucherType` VARCHAR(50) NOT NULL,
  `redeem_place` VARCHAR(255) NULL,

  `validFrom` DATE NOT NULL,
  `validUntil` DATE NOT NULL,
  `expires_at` DATE NULL,

  `conditions` TEXT NULL,
  `terms_conditions` TEXT NULL,

  `isUsed` BOOLEAN DEFAULT FALSE,
  `status` VARCHAR(255) DEFAULT 'unused',

  `created_at` TIMESTAMP NULL,
  `updated_at` TIMESTAMP NULL,
```

```

FOREIGN KEY (`customerID`) REFERENCES `customers`(`customerID`) ON
DELETE CASCADE,
FOREIGN KEY (`user_id`) REFERENCES `customers`(`customerID`) ON
DELETE CASCADE
);

```

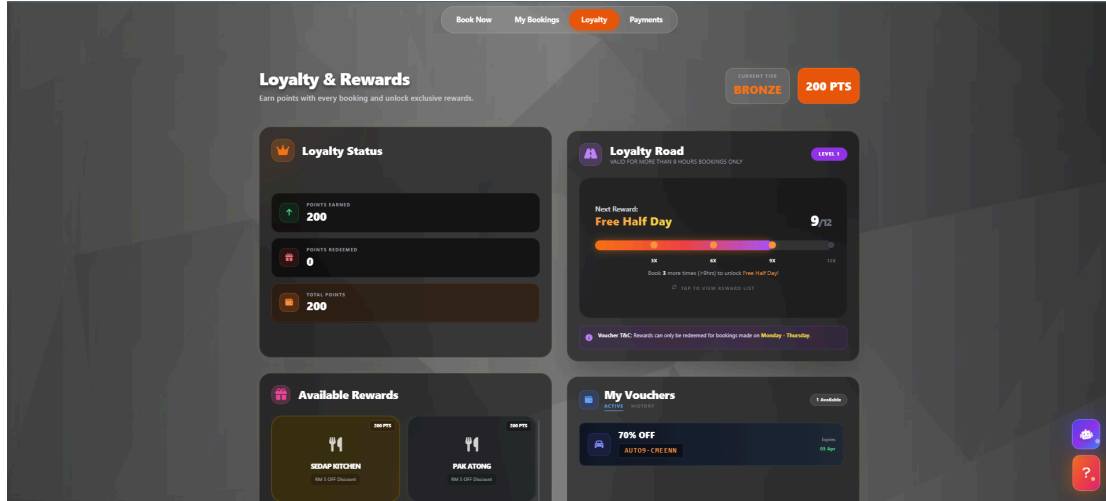


Figure 14: Loyalty & Rewards interface

The right bottom of the interface as shown in **Figure 14** is the voucher section. Customers will get a voucher through the loyalty road or could be given by the management side.

6.1.11 Loyalty & Reward Tables

```

CREATE TABLE `loyalty_points` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `user_id` BIGINT UNSIGNED NOT NULL,
  `points` INT DEFAULT 0,
  `tier` VARCHAR(255) DEFAULT 'bronze',
  `rental_bookings_count` INT DEFAULT 0,
  `created_at` TIMESTAMP NULL,
  `updated_at` TIMESTAMP NULL,
  FOREIGN KEY (`user_id`) REFERENCES `customers`(`customerID`) ON DELETE
  CASCADE
);

```

```

CREATE TABLE `loyalty_histories` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `user_id` BIGINT UNSIGNED NOT NULL,
  `points_change` INT NOT NULL,
  `reason` VARCHAR(255) NOT NULL,
  `created_at` TIMESTAMP NULL,
  `updated_at` TIMESTAMP NULL,

```

```
FOREIGN KEY (`user_id`) REFERENCES `customers`(`customerID`) ON DELETE
CASCADE
);
```

```
CREATE TABLE `rewards` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `name` VARCHAR(255) NOT NULL,
  `offer_description` VARCHAR(255) NOT NULL,
  `points_required` INT NOT NULL,
  `code_prefix` VARCHAR(255) NOT NULL,
  `validity_months` INT DEFAULT 3,
  `category` VARCHAR(255) DEFAULT 'Food',
  `icon_class` VARCHAR(255) DEFAULT 'fa-utensils',
  `color_class` VARCHAR(255) DEFAULT 'bg-yellow-600/20 border-yellow-500/30',
  `is_active` BOOLEAN DEFAULT TRUE,
  `milestone_step` INT NULL,
  `discount_percent` INT DEFAULT 0,

  `created_at` TIMESTAMP NULL,
  `updated_at` TIMESTAMP NULL
);
```

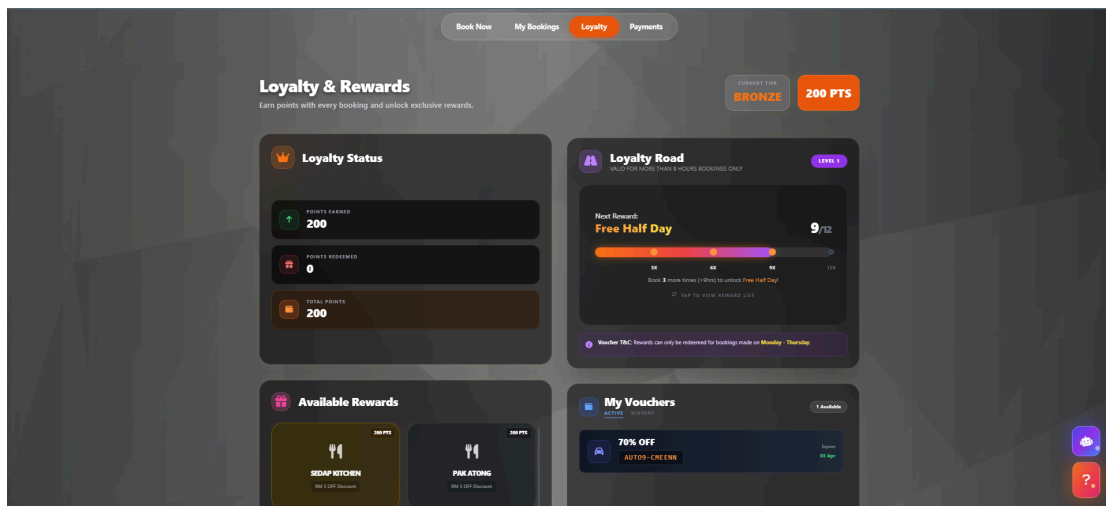


Figure 14: Loyalty & Rewards interface

Figure 14 also included the loyalty feature for the system. Customers could get the loyalty points through every booking and use those points to claim for the rewards.

6.1.12 Vehicle Inspections Table

```
CREATE TABLE `inspections` (
  `inspectionID` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `bookingID` BIGINT UNSIGNED NOT NULL,
```

```

`staffID` BIGINT UNSIGNED NULL,
`inspectionType` VARCHAR(50) NOT NULL,
`inspectionDate` DATETIME NOT NULL,
`damageCosts` DECIMAL(10, 2) DEFAULT 0.00,
`photosBefore` TEXT NULL,
`photosAfter` TEXT NULL,
`fuelBefore` VARCHAR(50) NULL,
`fuelAfter` VARCHAR(50) NULL,
`mileageBefore` INT NULL,
`mileageAfter` INT NULL,
`created_at` TIMESTAMP NULL,
`updated_at` TIMESTAMP NULL,
FOREIGN KEY (`bookingID`) REFERENCES `bookings`(`bookingID`) ON
DELETE CASCADE,
FOREIGN KEY (`staffID`) REFERENCES `staff`(`staffID`)
);

```

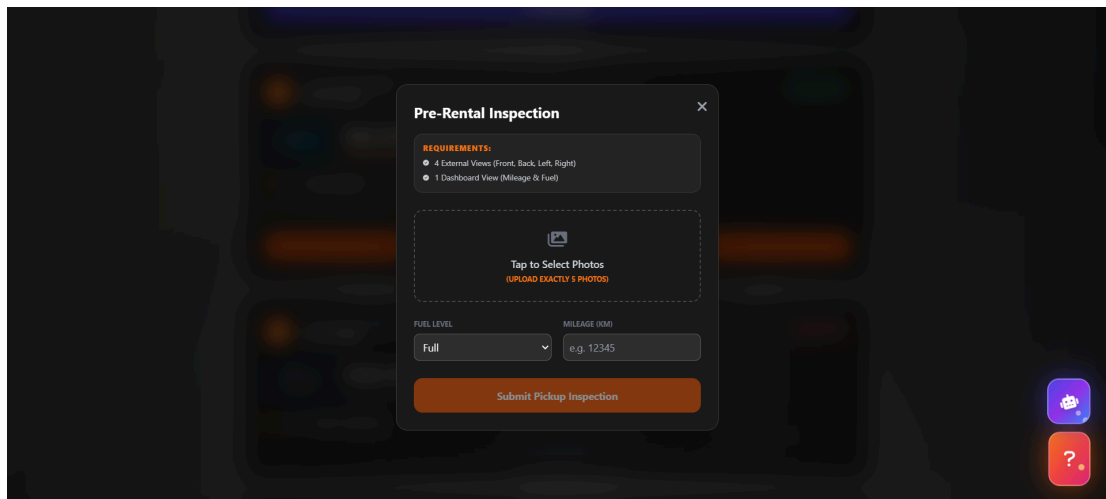


Figure 15: Vehicle Inspection (Customer)

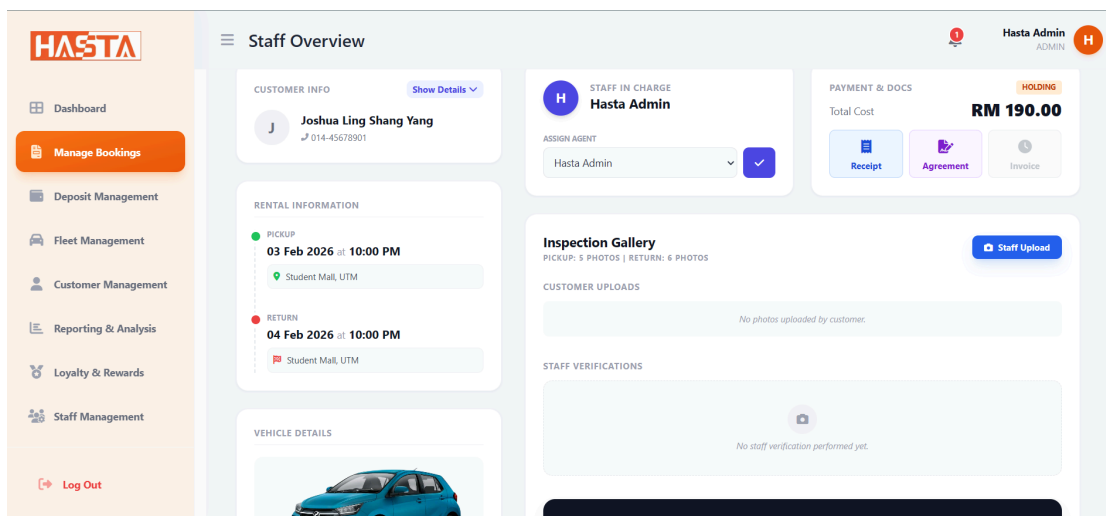


Figure 16: Vehicle Inspection (Staff)

In both vehicle inspection interfaces in **Figure 15** and **Figure 16**, Customer is required to upload the vehicle inspection pictures before and after the rental for references while staff could choose to upload the photo during the inspection phase if there is any error of information.

6.1.13 Feedbacks Table

```
CREATE TABLE `feedbacks` (  
  `feedbackID` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
  `customerID` BIGINT UNSIGNED NOT NULL,  
  `bookingID` BIGINT UNSIGNED NULL,  
  `rating` INT NOT NULL,  
  `comment` TEXT NULL,  
  `type` VARCHAR(50) NOT NULL,  
  `status` VARCHAR(50) DEFAULT 'Pending',  
  `adminNotes` TEXT NULL,  
  `created_at` TIMESTAMP NULL,  
  `updated_at` TIMESTAMP NULL,  
  FOREIGN KEY (`customerID`) REFERENCES `customers`(`customerID`) ON  
DELETE CASCADE,  
  FOREIGN KEY (`bookingID`) REFERENCES `bookings`(`bookingID`) ON  
DELETE SET NULL  
);
```

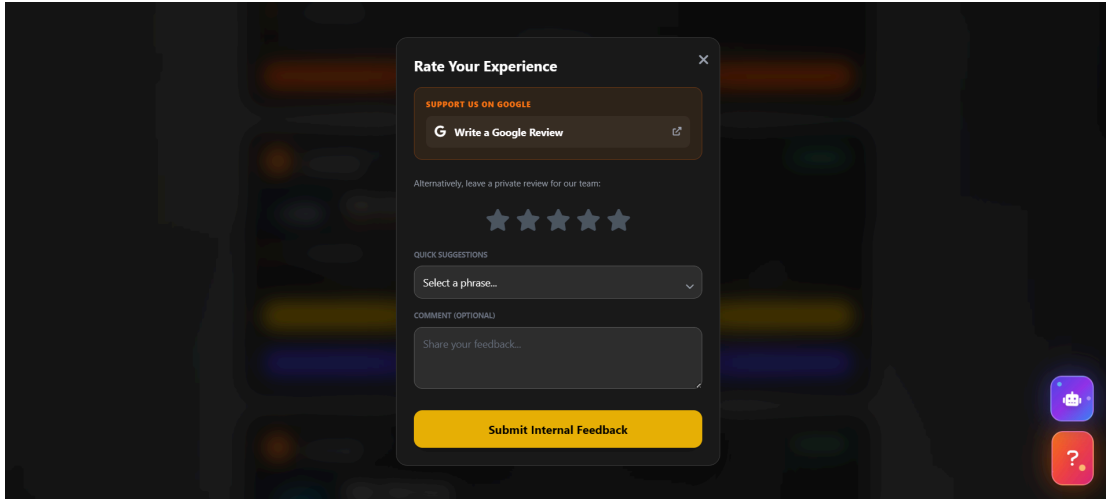


Figure 17: Feedback interface

Figure 17 shows the feedback could be done by the customer after the rental. Customers can rate the entire booking and give the suggestions to the management side for improvements.

6.2 SQL Statement (DML)

6.2.1 User & Staff Module DML

These operations manage customer registration, profile updates and staff administrative actions.

Register a New Customer:

```
INSERT INTO CUSTOMERS (fullName, email, phoneNo, password, faculty,  
nationality, dob)  
VALUES ('Ajmi User', 'ajmi@example.com', '0123456789', 'hashed_password', 'FKE',  
        'Malaysian', '2000-01-01');
```

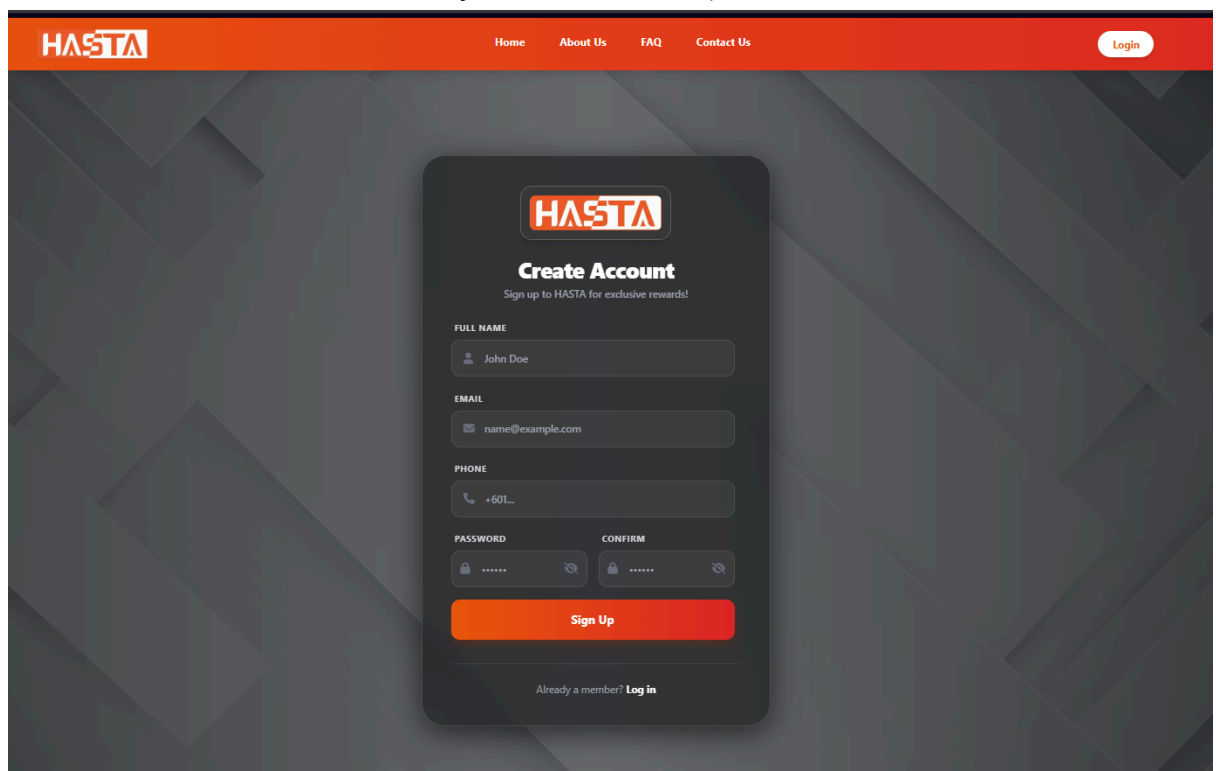


Figure 18: Register New Customer Interface

Figure 18 represents the entry point for new users to join the HASTA platform. Its primary function is to collect and verify the necessary information to create a secure user profile.

Update Customer Blacklist Status:

```
UPDATE CUSTOMERS  
SET blacklisted = 1, blacklist_reason = 'Multiple late returns'  
WHERE customerID = 101;
```

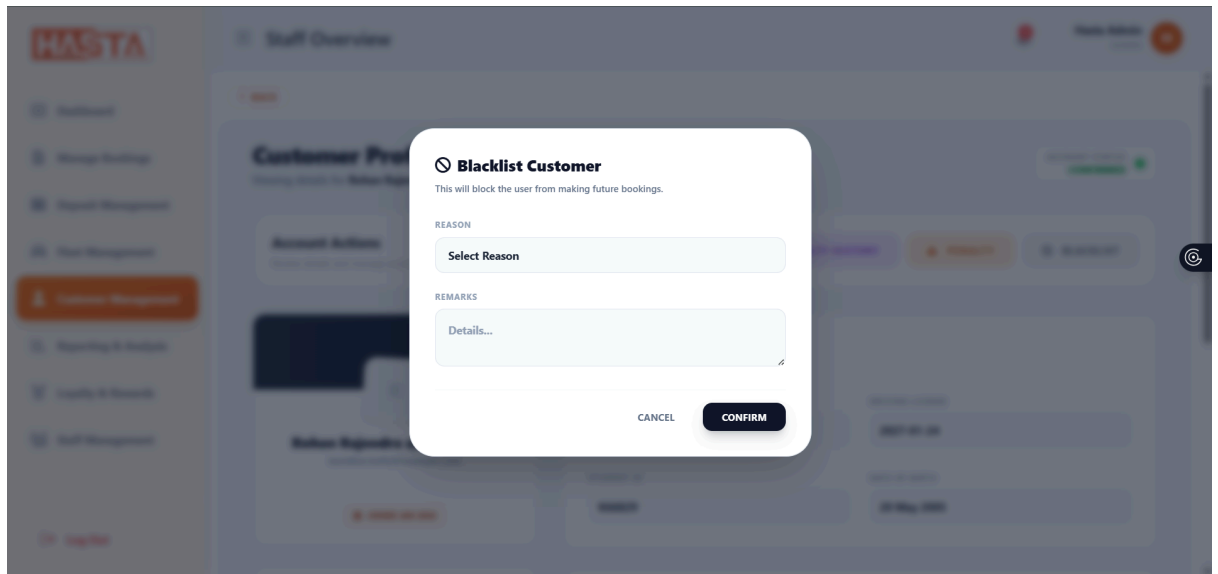



Figure 19: Customer Blacklist Update Interface

Figure 19 depicts the Customer Blacklist Update Interface, which serves as an administrative control to restrict specific users from accessing the system's services.

Add Emergency Contact:

```
INSERT INTO EMERGENCY_CONTACTS (customerID, contact_name,
contact_no)
VALUES (101, 'Emergency Contact A', '0119876543');
```

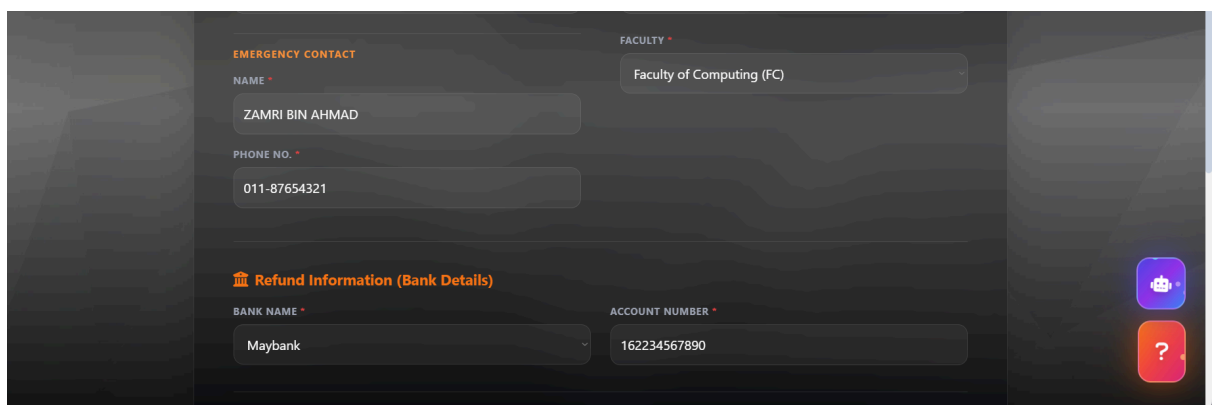


Figure 20: Add Emergency Contact Interface

Figure 20 illustrates the Add Emergency Contact Interface, which captures critical safety and financial information for the customer profile.

6.2.2 Booking & Operations Module DM

These manage the life cycle of a rental, from vehicle availability to inspections.

Insert a New Booking:

```
INSERT INTO BOOKINGS (customerID, vehicleID, bookingDate, returnDate,  
pickupLocation, totalCost, bookingStatus)  
VALUES (101, 5, '2026-02-05', '2026-02-07', 'Arked Angkasa', 150.00, 'Pending');
```

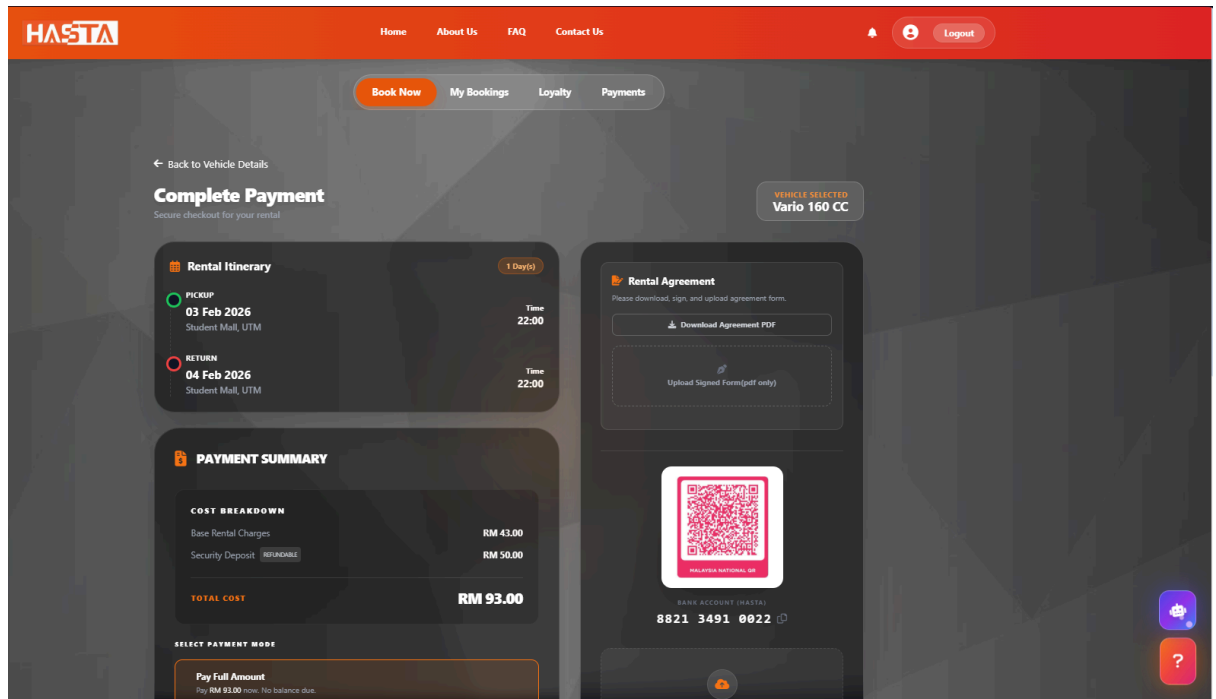


Figure 21: Insert New Booking Interface

Figure 21 illustrates the Insert New Booking Interface, which serves as the final checkout and confirmation stage for a vehicle rental.

Set Vehicle Availability:

UPDATE VEHICLES

SET availability = 0

WHERE vehicleID = 5;

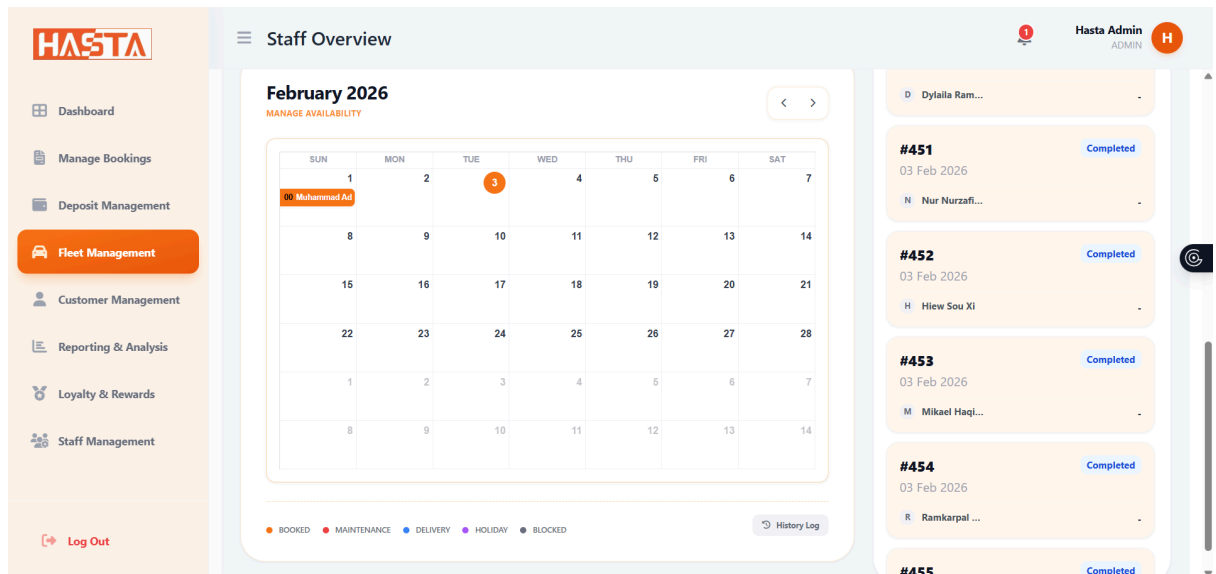


Figure 22: Set Vehicle Availability

Figure 22 presents the **Set Vehicle Availability Interface**, which provides a centralized calendar view for administrators to manage the operational status of the fleet.

Record a Maintenance Task:

INSERT INTO MAINTENANCES (vehicleID, staffID, type, date, description, cost)
VALUES (5, 1, 'Oil Change', '2026-02-01', 'Scheduled maintenance', 120.00);

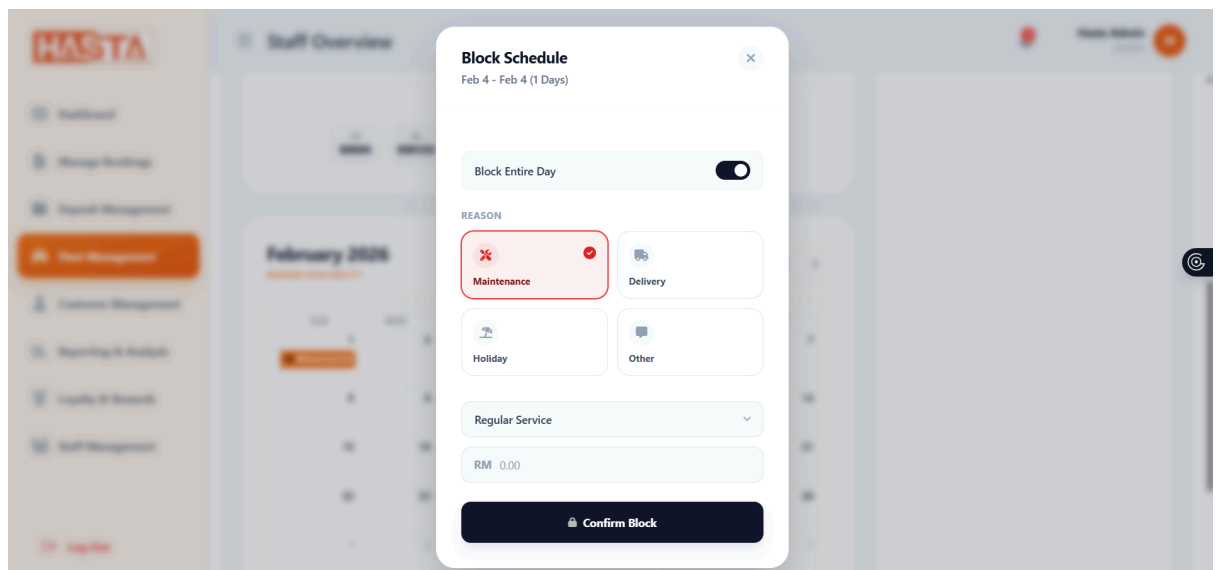


Figure 23: Record Maintenance Task Interface

Figure 23 illustrates the **Record Maintenance Task Interface**, which serves as the primary tool for managing vehicle health and operational downtime.

6.2.3 Finance & Loyalty Module DML

These handle payments, penalties and the reward system tracking.

Record a Payment:

```
INSERT INTO PAYMENTS (bookingID, amount, transactionDate, paymentMethod,
paymentStatus)
```

```
VALUES (501, 150.00, NOW(), 'Online Transfer', 'Success');
```

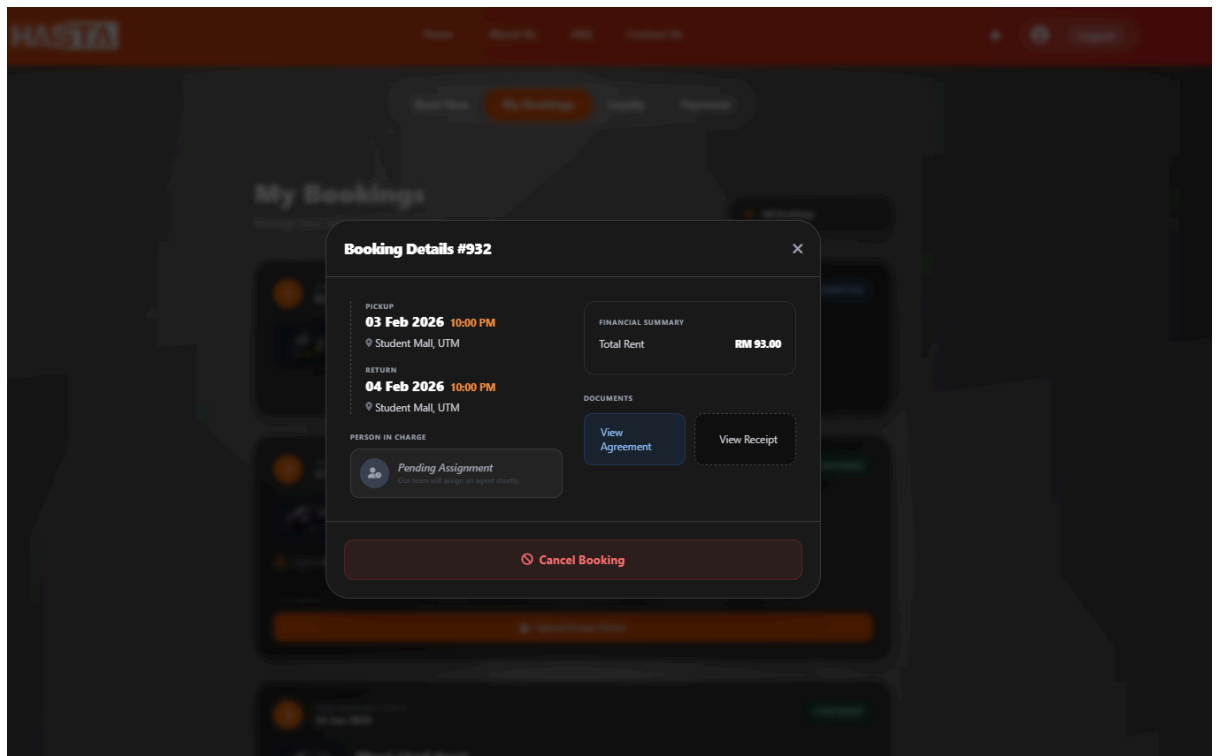


Figure 24: Record Payment Interface

Figure 24 illustrates the **Record Payment Interface**, which acts as a centralized dashboard for a specific booking to ensure the user has complete oversight of their rental transaction.

Impose a Penalty:

```
INSERT INTO PENALTIES (customerID, bookingID, amount, reason, status)
VALUES (101, 501, 50.00, 'Late return of 2 hours', 'Unpaid');
```

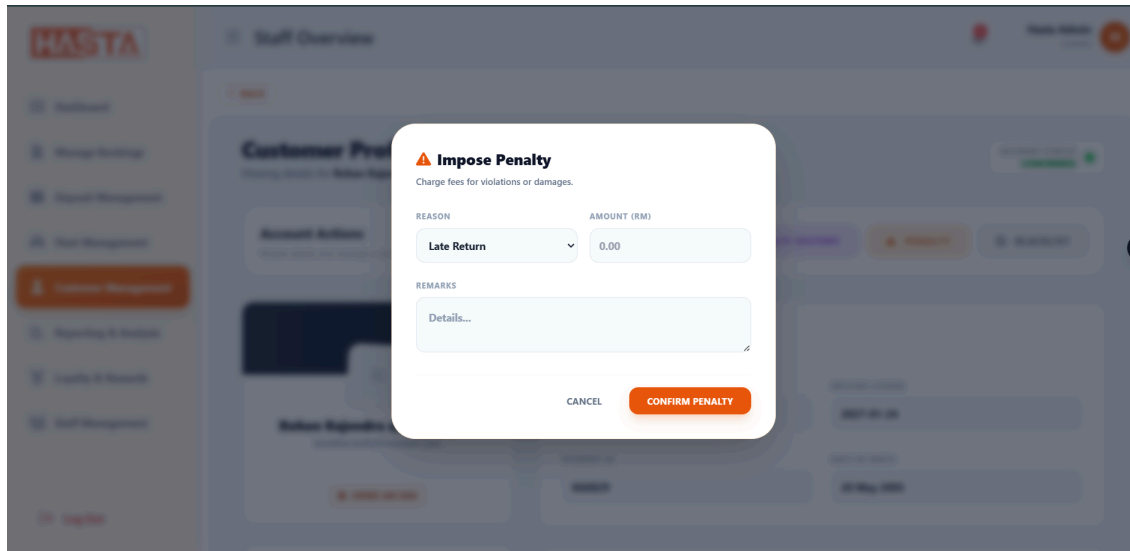


Figure 25: Impose Penalty Interface

Figure 25 shows Impose Penalty Interface where the administrative tool simplifies the management of violations or damages through a standardized workflow.

Award Loyalty Points:

UPDATE LOYALTY_POINTS

SET points = points + 10, rental_bookings_count = rental_bookings_count + 1

WHERE user_id = 101;

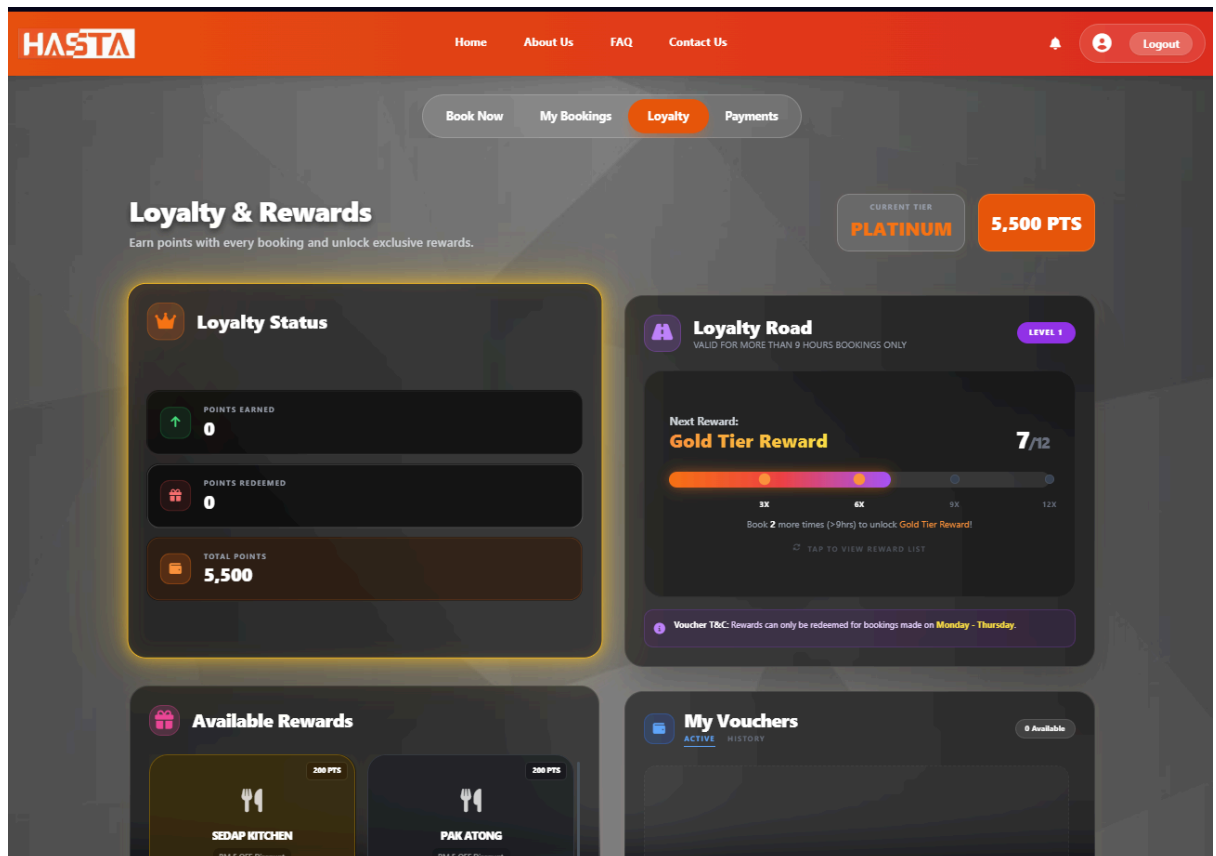


Figure 26: Award Loyalty Point Interface

Figure 26 illustrates the **Award Loyalty Point Interface**, which serves as a comprehensive dashboard for users to track their engagement and redeem earned benefits within the HASTA system.

7. Summary

This Logical Design Report outlines a comprehensive digital transformation strategy for Hasta Travel & Tours company. Currently, the company is highly relying on a manual system of conducting business through WhatsApp, which is prone to errors and security risks. The proposed digital transformation project aims to introduce a specialized Vehicle Rental Management System. By embracing a digital system of conducting business, the company will seek to address problems such as double bookings, as well as seeking a better way to improve customer experience through a more professional system.

A critical aspect of this project is a technical design with a focus on data integrity and efficiency. The design is based on a Logical ERD and a Data Dictionary that identify entities within the system, which include the staff, customers, vehicles, bookings and payments. In addition, for the database to be effective and non-redundant, we have to do a normalization process up to 3NF. In the result, a clean relational schema will be performed. In fundamental features of a vehicle rental agency, there is a need for a sophisticated business logic that is integrated into this system. It has included a multi-tiered loyalty points program for its clients and penalty actions for late returns and damages.

Apart from that, this report also identifies an in-depth analysis of the system's logic through a Use Case and Activity Diagram. These two diagrams identify all the activities of a user from searching for a vehicle to paying for the vehicle and getting a refund for the deposit. Furthermore, SQL code for developing this system is also included in this report. It is a comprehensive blueprint for developing a scalable and user-friendly management system that meets the objectives that are requested by the clients.

8. Appendices

8.1. Use Case Diagram for the system

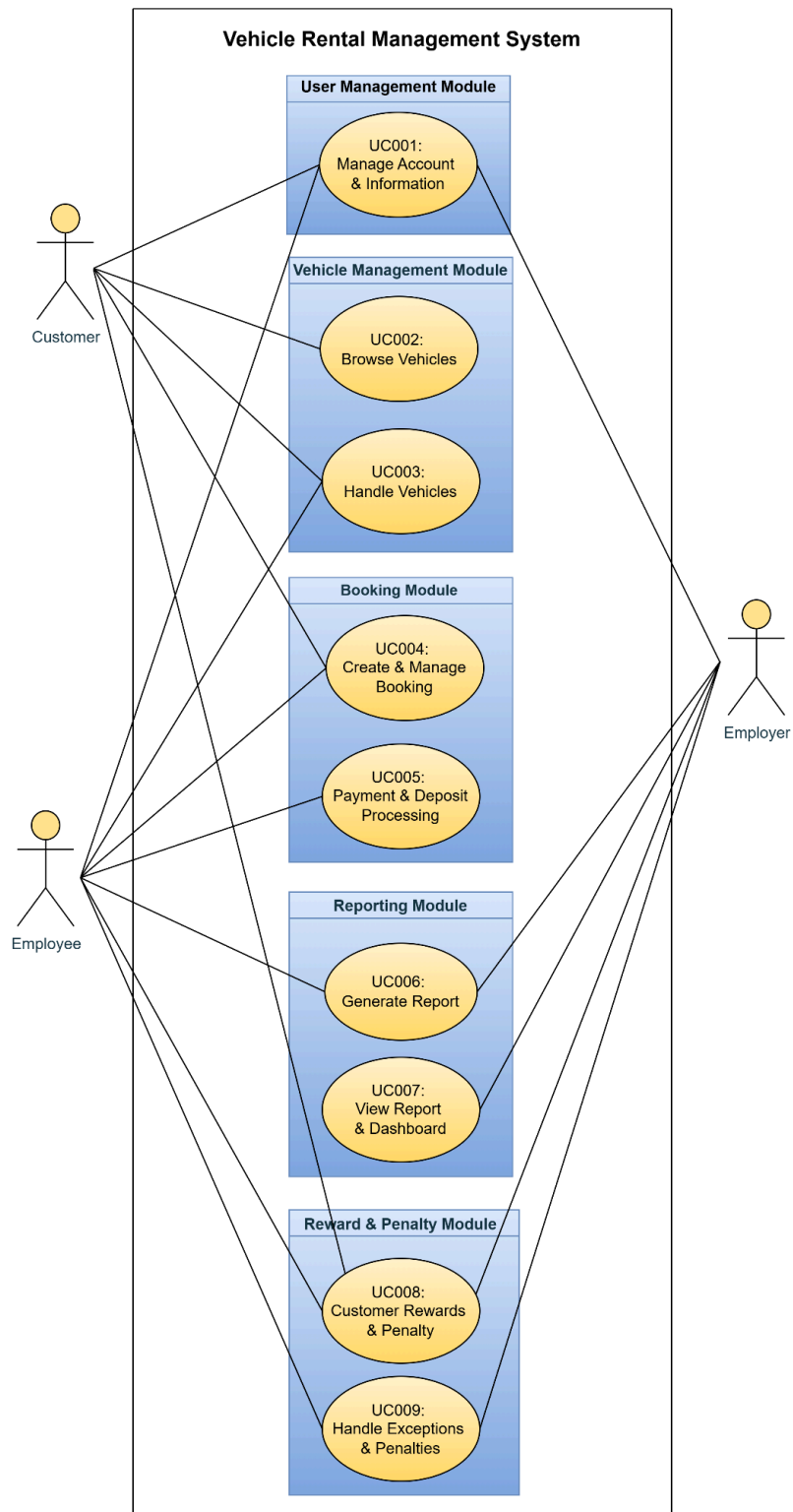


Figure 27: Use Case Diagram for Vehicle Rental Management System

8.2. Activity Diagram for every use case identified

8.2.1. UC001

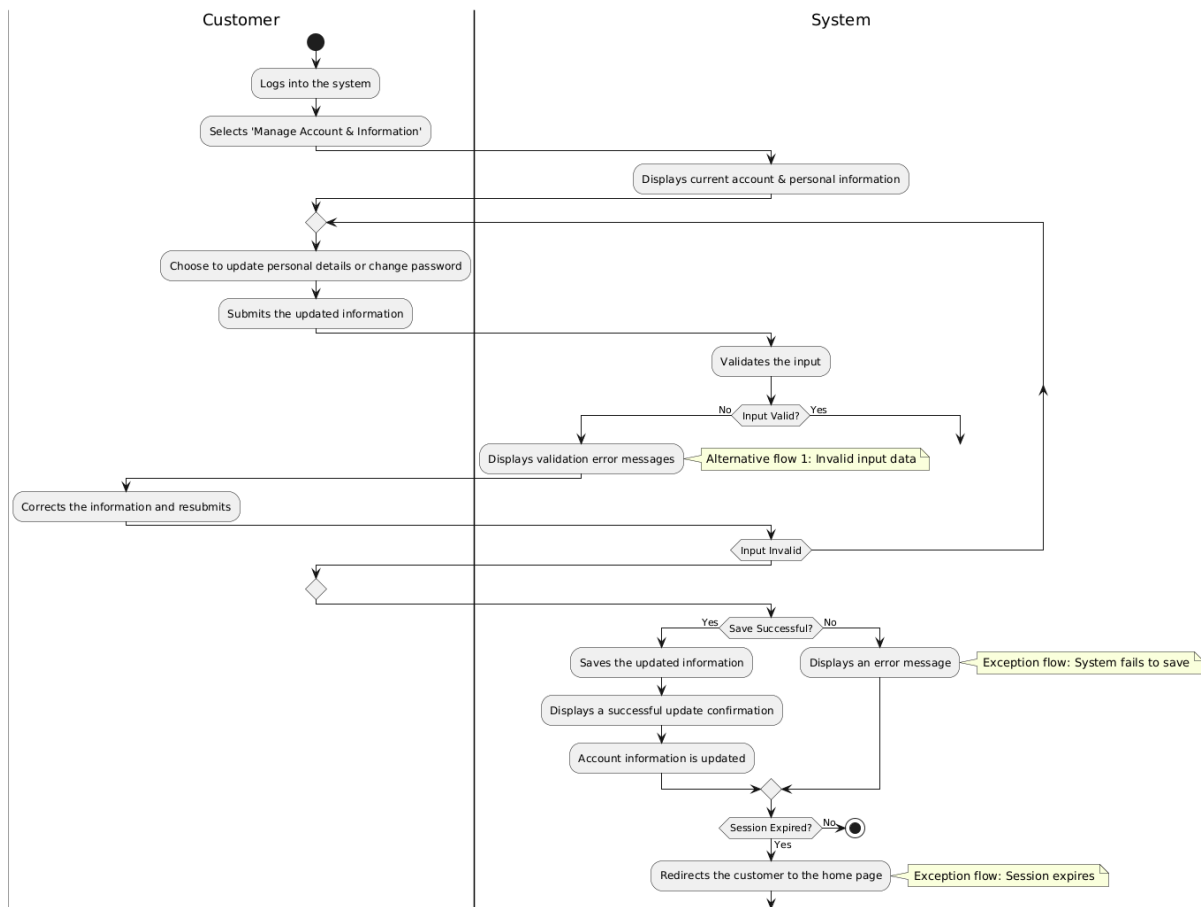


Figure 28: Activity Diagram for Manage Account & Information

Figure 28 details the chronological interactions between the Customer, System and Database required to update account information. It highlights the sequential steps of retrieving current data, validating user inputs and confirming the update in the database while handling potential exceptions.

8.2.2. UC002

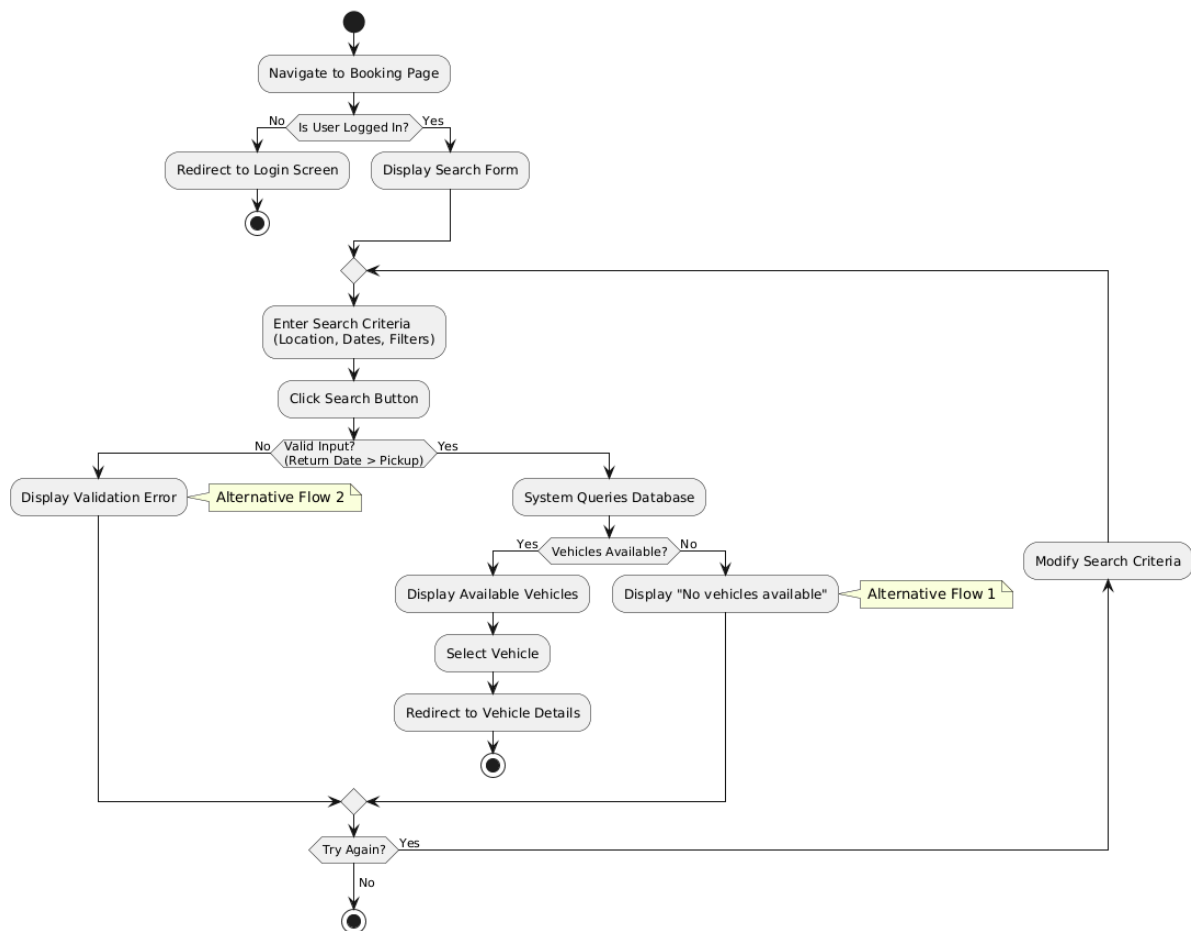


Figure 29: Activity Diagram for Browse Vehicles

Figure 29 describes the activity workflow for browsing cars, which starts with a log in prior to gaining access to the searching interface. Successful completion of this activity requires that a user chooses an available car from among the list created and is then taken to another webpage for details.

8.2.3. UC003

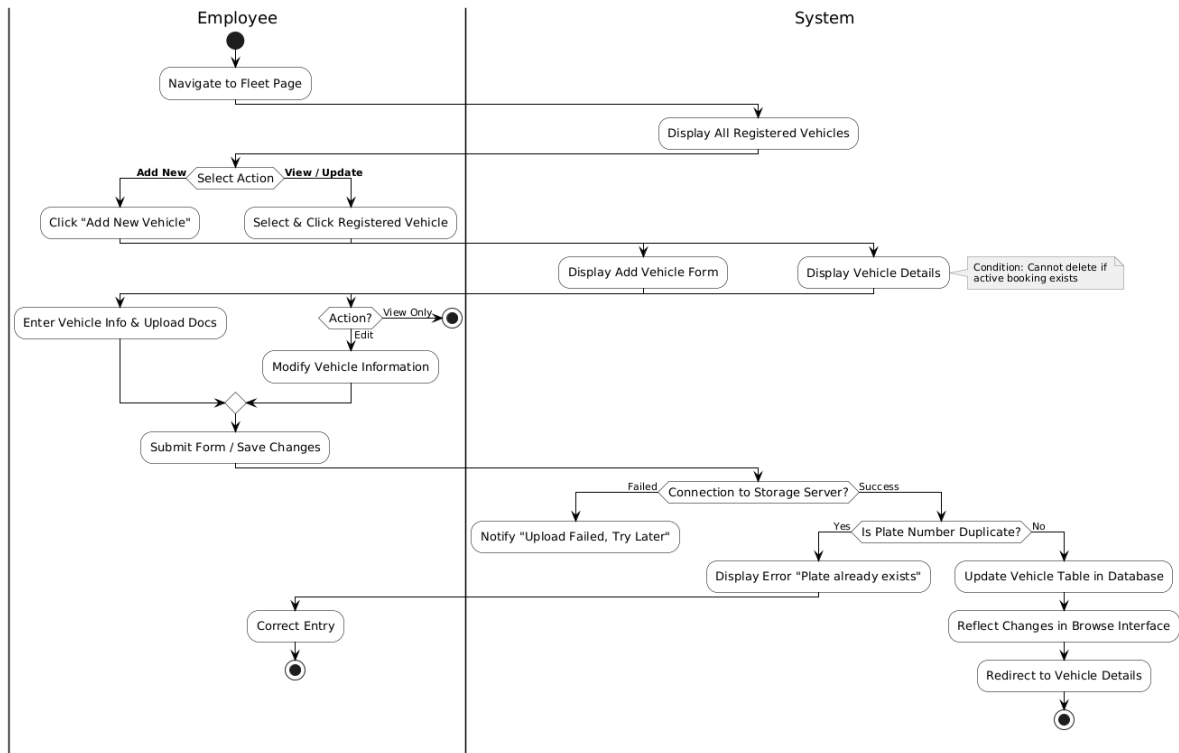


Figure 30: Activity Diagram for Handle Vehicles

Based on **Figure 30**, it shows system interaction that clearly details the message sequence flow from the Staff, Fleet UI, Vehicle Controller, to the Database. The system has broken the process into two different operation flows, Main Flow, which deals with the display and update of current data in vehicles, while Alternative Flow 1 deals with adding new inventory to the system. Error handling is also shown in the system interaction through alternative flows.

8.2.4. UC004

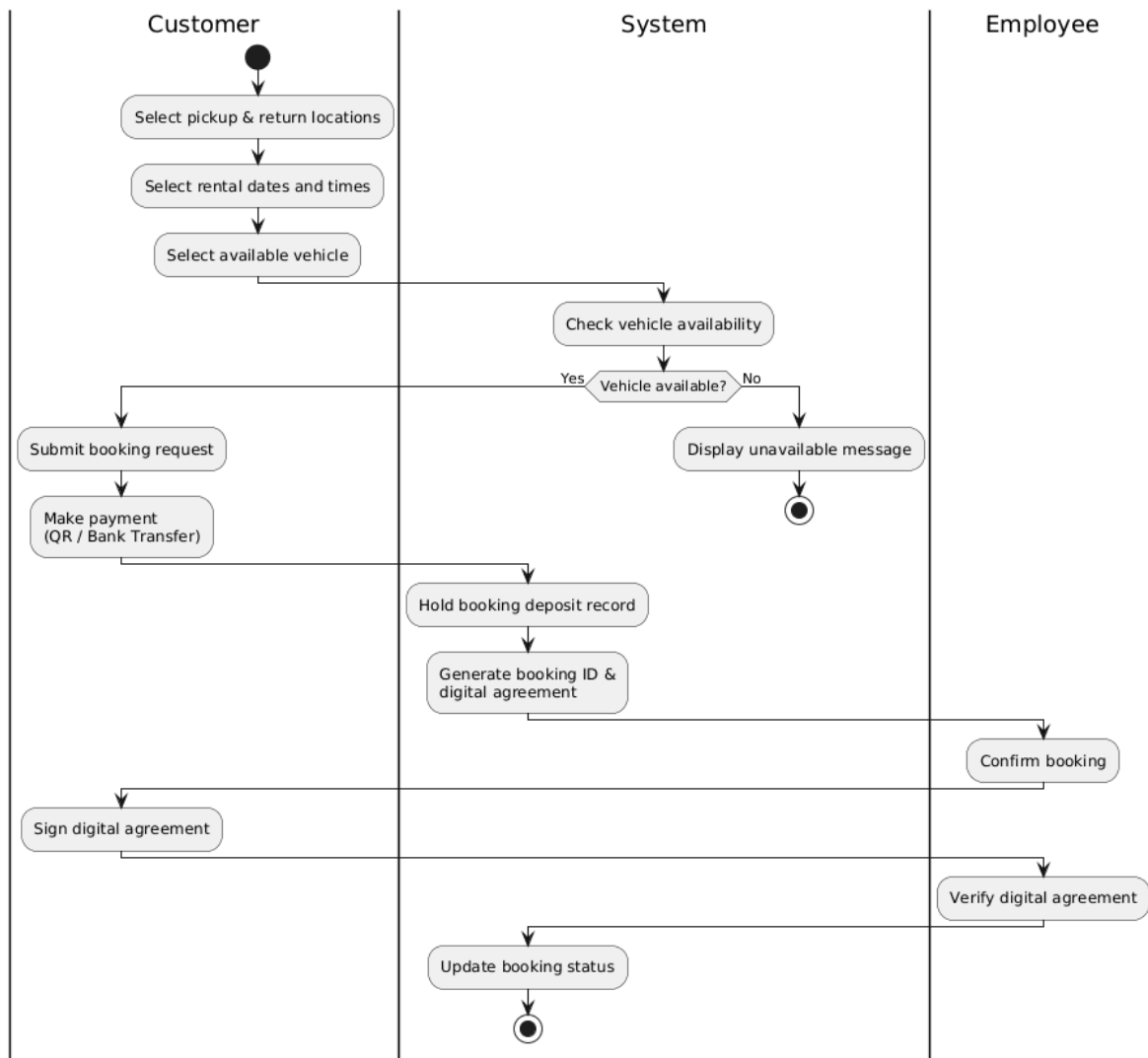


Figure 31: Activity Diagram for Manage Account & Information

Figure 31 illustrates a sequence diagram of the customer's booking process. The customer interacts with BookingPage and BookingSystem while The Employee interacts with PaymentSystem. The system will display unavailable message if the vehicle is not available.

8.2.5. UC005

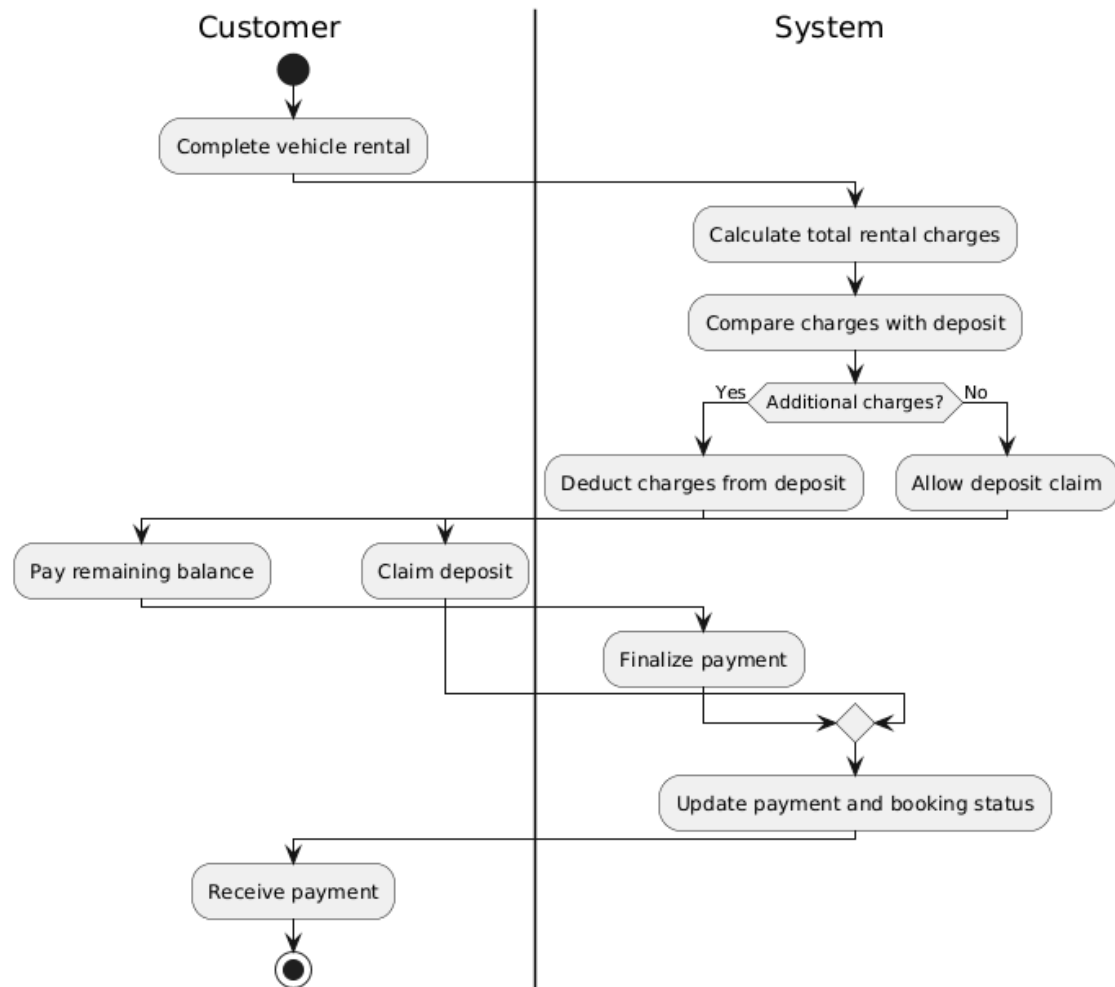


Figure 32: Activity Diagram for Payment & Deposit Processing

Figure 32 illustrates the sequence diagram for the payment processing process happened between Payment System and Booking System. The process begin when complete vehicle rental and the booking system calculate total rental charges with deposit held if there is problem with the booking the system will deduct charges from deposit and request for remaining balance, if there is no problem with the booking the system will allow deposit claims.

8.2.6. UC006

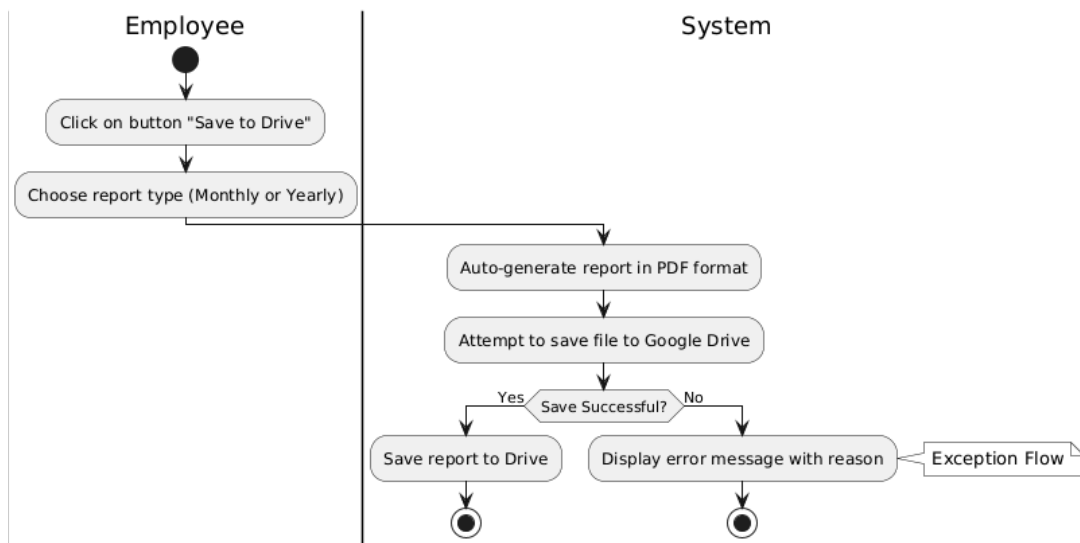


Figure 33: Activity Diagram for Generate Report

Figure 33 illustrates the employee's workflow for creating reports. It shows the process of selecting a report type (monthly or yearly) and the system's automated actions to generate the PDF and save it directly to Google Drive.

8.2.7. UC007

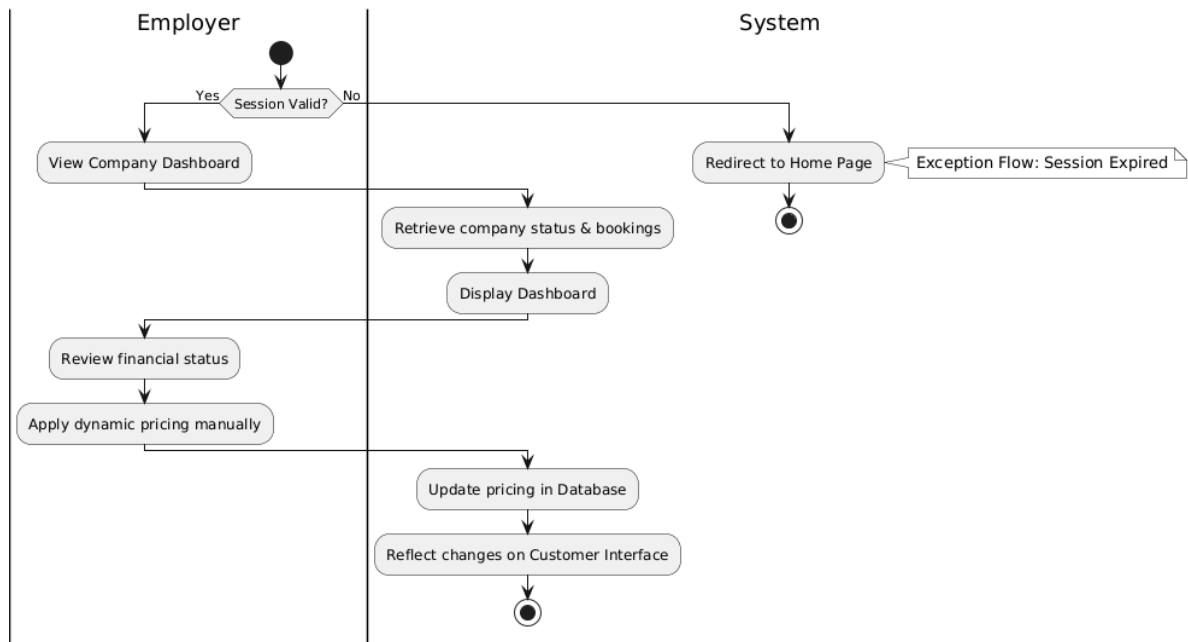


Figure 34: Activity Diagram for View Review & Dashboard

Figure 34 maps the employer's interaction with the company dashboard. It visualizes the flow of viewing real-time financial status and bookings, as well as the logic for manually applying dynamic pricing updates to the system.

8.2.8. UC008

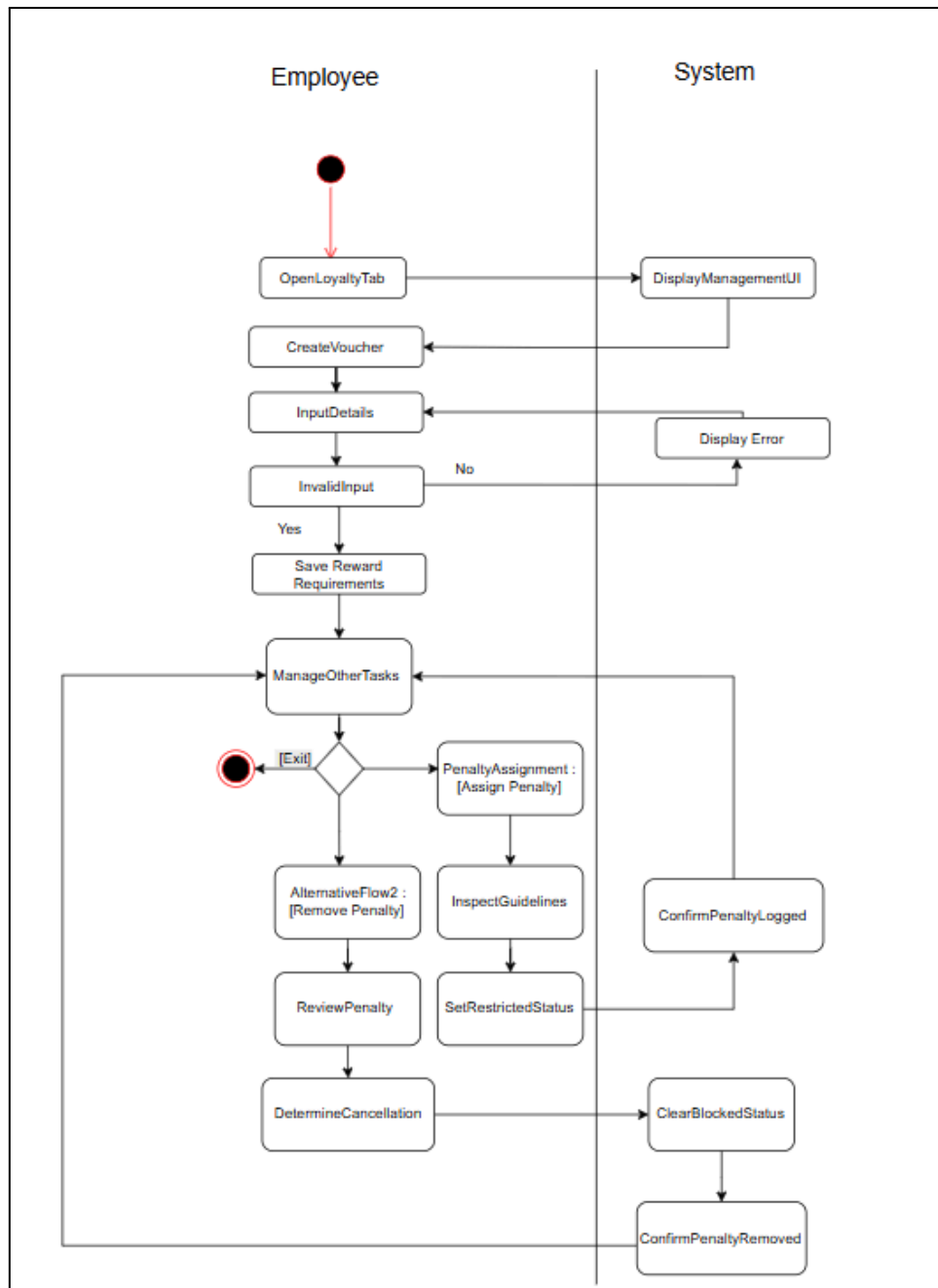


Figure 35: Activity Diagram for Customer Rewards & Penalty

Figure 35 illustrates the flow of exchange of data required to manage loyalty and penalty features within the dashboard. It depicts the Employee initiating tasks like voucher creation while the System validates inputs and updates the backend status. The workflow specifically highlights the transition of saving reward requirements where the System subsequently logs or clears penalties and updates user restrictions based on the Employee's real-time decisions.

8.2.9. UC009

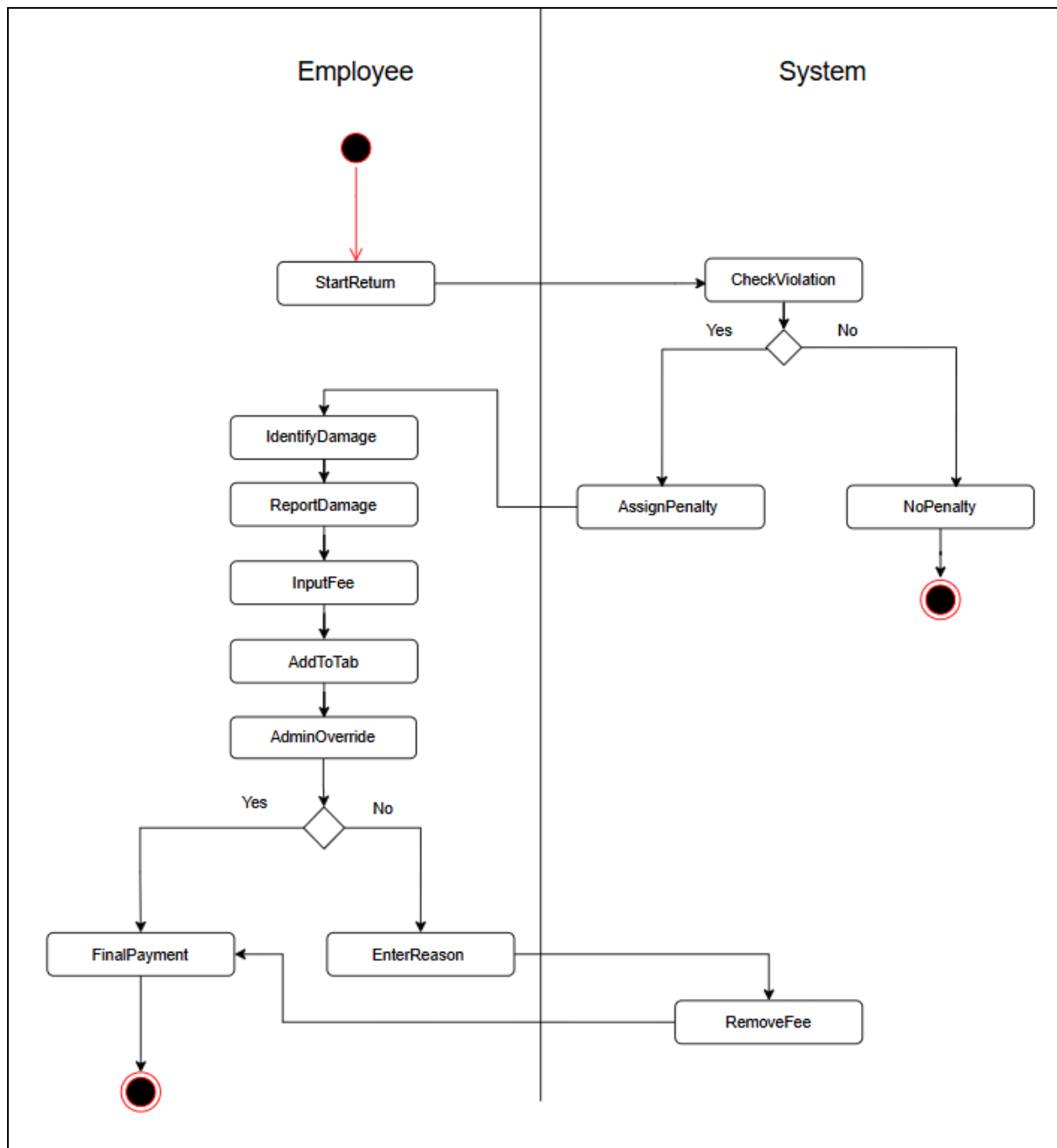
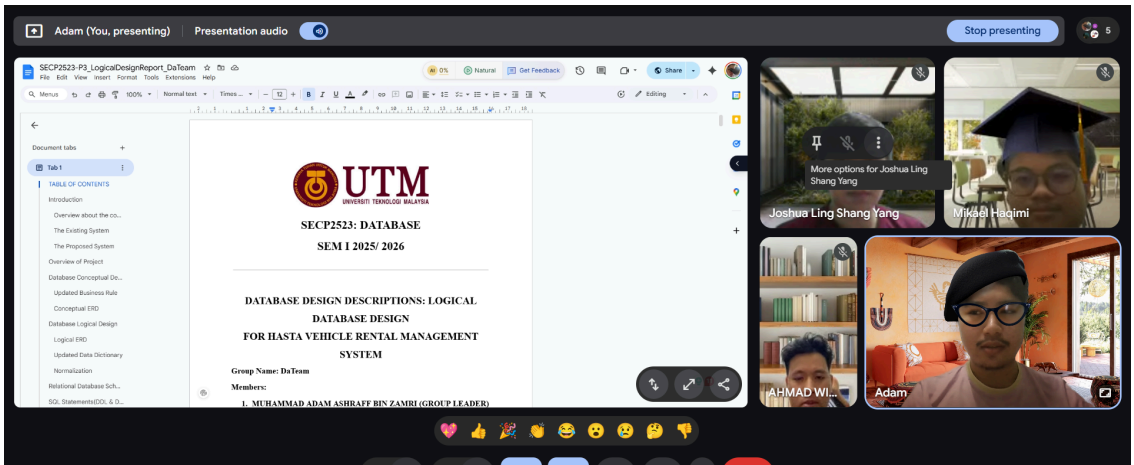


Figure 36: Activity Diagram for Handle Exceptions & Penalties

Figure 36 illustrates the chronological exchange of data required to handle vehicle returns and manage potential damage charges. It depicts the **System** executing an automated violation check, which subsequently prompts the **Employee** to assess the vehicle for damages and input applicable fees. The diagram also captures the update process during an **AdminOverride**, where the System records a justification for removing fees before the Employee completes the final payment and closes the return process.

8.3. Meeting Log

Meeting Minute #1

DATE/TIME	26 th January 2026		
LOCATION	Google Meet(Online Meeting)		
AGENDA	Task Distribution and Finalize Report		
MEETING MC	MUHAMMAD ADAM ASHRAFF BIN ZAMRI		
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
MUHAMMAD ADAM ASHRAFF BIN ZAMRI	12.30 PM	-	
MIKAEL HAQIMI BIN NAHAR JUNAIDI	12.30 PM	-	
AHMAD WILDAN BIN MAZANI	12.30 PM	-	
JOSHUA LING SHANG YANG	12.30 PM	-	
GROUP PHOTO			
			
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTION AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Task Distribution	Adam distributes all tasks evenly.	Adam

2	Task Discussion	All members read and identify the requirements for this task.	All members
3	Finalize Reports	All members recheck all information and discuss possible improvements.	All members
4	Meeting ended	Adam ended the meeting at 4.30 PM	Adam

Meeting Minute #2

DATE/TIME	2 nd January 2026	
LOCATION	MA1	
AGENDA	Finalize Report	
MEETING MC	MUHAMMAD ADAM ASHRAFF BIN ZAMRI	
ATTENDANCE		
NAME	TIME	REASON FOR ABSENCE
MUHAMMAD ADAM ASHRAFF BIN ZAMRI	1.30 PM	-
MIKAEL HAQIMI BIN NAHAR JUNAIDI	1.30 PM	-
AHMAD WILDAN BIN MAZANI	1.30 PM	-
JOSHUA LING SHANG YANG	1.30 PM	-
GROUP PHOTO		



MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTION AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Finalize Reports	All members recheck all information and discuss possible improvements.	All members
2	Meeting ended	Adam ended the meeting at 5.30 PM	Adam